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DEDICATION

This work is proudly dedicated to:

Our beloved parents, the anchors of our lives, whose unconditional love, endless sacrifices, and continuous prayers have been the cornerstone of our strength and the light guiding us toward this achievement.

Our esteemed supervisor, Dr. Mohammad Alsharo whose profound wisdom, inspiring mentorship, and unwavering support illuminated our academic path and pushed us to strive for excellence.

We dedicate the fruits of this labor to you.

ACKNOWLEDGMENTS

الحمد لله حمدًا كثيرًا طيبًا مباركًا فيه.

Our deepest appreciation and heartfelt thanks go to our parents. We are forever grateful for your endless support, patience, and unwavering belief in us. This milestone would not have been possible without your sacrifices.

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We also extend our sincere thanks to the members of our examination committee, Dr. Anas Jebreen and Dr. Manar Mizher, for their valuable time, insightful feedback, and for evaluating our work.

Finally, we thank our siblings, friends, and colleagues who encouraged us along the way and contributed to the completion of this milestone.

Committee Decision

Examination committee	Signature
1. ……	……
2. ……	……
3. ……	……
4. ……	……

This project was submitted in partial fulfillment of the requirements for the Bachelor's degree in Computer Information Systems at the Faculty of Prince Al-Hussein Bin Abdullah II for Information Technology, Al al-Bayt University, Second Semester 2025/2026.

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Abstract

CareerCompass is an interactive web-based academic guidance platform developed by Al al-Bayt University students to address the critical challenge of university major selection faced by high school students in Jordan. The platform serves as a comprehensive digital advisor, bridging the gap between a student's academic potential and future career opportunities through data-driven decision-making. The system utilizes a dual-track logic to cater to two distinct groups: Pre-Tawjihi students (Grades 9–11), who receive actionable improvement plans, and Tawjihi students (Grade 12), who receive immediate university matching. CareerCompass combines an extensive database of local university majors with an intelligent interest assessment test that evaluates students across multiple dimensions. Furthermore, it includes a comprehensive university exploration tool allowing students to review majors based on required skills, study duration, and career opportunities. Technically, the platform features a user-friendly interface built as a Single Page Application (SPA) with modern web technologies (Vue.js 3 and Tailwind CSS), backed by a robust server-side infrastructure using Python (FastAPI) and SQLite. While currently relying on advanced programmed logic, the system architecture lays the foundation for future Artificial Intelligence (AI) integration. Ultimately, CareerCompass aims to reduce the likelihood of unsuitable major choices, enhance academic satisfaction, and align with Jordan's vision for digital transformation in higher education, empowering youth to make confident, informed career choices.

Chapter 1: Overview of the Project

1.1 Introduction Education is the cornerstone of personal and national development. The transition from high school to university represents one of the most critical decision points in a student's academic journey, shaping not only their next four years of study but also their future career trajectory, earning potential, and overall life satisfaction. In Jordan, this juncture often causes significant stress and confusion, as this foundational choice is frequently made without adequate guidance or reliable information.

In the current educational landscape, students face several persistent challenges that hinder informed decision-making:

- **Information Scarcity:** Comprehensive, reliable information about different majors is often scattered across multiple sources or unavailable.
- **Lack of Personalization:** Generic academic advising often fails to account for individual strengths, interests, and career aspirations.
- **Family Influence:** Students frequently rely on family opinions or social pressure rather than data-driven insights about their own preferences.
- **Comparison Difficulty:** Evaluating multiple majors side-by-side based on meaningful criteria is fragmented and time-consuming.
- **Poor Decision-Making:** The absence of effective pre-enrollment guidance contributes to high rates of major-switching, which wastes valuable time and resources while negatively impacting student satisfaction.

With universities offering an increasingly diverse array of majors, students need a systematic, intelligent tool to navigate these options. With the advancement of web technologies, digital platforms can now offer the personalized guidance that was previously inaccessible. CareerCompass is designed to address these challenges by creating a centralized, interactive platform. It serves as a digital mentor that combines comprehensive

major information, intelligent assessment capabilities, and personalized guidance, effectively aligning a student's academic reality with their personal and professional goals.

1.2 Problem Statement The project addresses several critical issues facing the Jordanian educational sector, where the absence of data-driven guidance has led to a disconnect between student potential and academic reality. The key problems that CareerCompass aims to solve are:

- **Lack of Accessible and Organized Information:** High school students struggle to find comprehensive, reliable information about university majors available in Jordan. Critical details are often scattered across multiple university websites, outdated sources, and unverified forums, making it difficult for students to compare majors or understand career pathways systematically.
- **Absence of Effective Academic Guidance Systems:** Pre-enrollment academic advising is limited or non-existent in many educational contexts. Consequently, students often rely on family opinions or social pressure rather than data-driven insights about their own abilities. Currently, no accessible mechanism exists to scientifically match students with majors based on their personality traits and cognitive preferences.
- **Confusion Due to Major Proliferation:** The large number of available majors (science, engineering, arts, business, etc.) creates decision paralysis and anxiety, particularly during the high-pressure Tawjihi year. Students lack a structured way to narrow down options or understand the distinctions between similar programs, and career outcomes for different majors are often not clearly communicated.
- **Mismatch Between Student Preferences and Chosen Major:** Many students do not have clarity on their own interests, strengths, and career goals before making a major choice. Without objective self-assessment tools, students cannot identify their

academic and professional preferences accurately, leading to dissatisfaction and poor academic performance.

- **High Rate of Major-Switching:** A significant percentage of university students change their major after enrollment due to early poor decision-making. This leads to academic delays, extended time-to-degree, and a waste of financial and educational resources for both families and institutions.
- **Lack of Early Planning (Grades 9-11):** Younger students often reach the critical Tawjihi stage without a clear goal or knowledge of the prerequisite skills needed for their desired fields. The current system lacks tools to help pre-Tawjihi students identify skill gaps early enough to improve them before applying to universities.

1.3 Proposed Solution CareerCompass proposes a comprehensive, integrated web-based platform that serves as an intelligent digital academic advisor. The solution addresses the identified problems by automating the counseling process through a robust, logic-driven architecture. The core components of the solution include:

1. **Dual-Track Logic System:** The platform offers tailored workflows to serve distinct user groups:
    - **Pre-Tawjihi Track (Grades 9-11):** Focuses on early planning by generating custom Action Plans that identify skill gaps and suggest specific subject improvements to reach future goals.
    - **Tawjihi Track (Grade 12):** Focuses on immediate decision-making by providing direct university matching based on final stream selection.
2. **Comprehensive & Localized Major Database:** Unlike generic lists, this component provides a detailed repository aligned strictly with the updated Jordanian secondary fields (**Health, Engineering & IT, Humanities, and Business**). For each major, the system provides:
    - Program overviews, curriculum structures, and course requirements.

    - Required foundational skills and typical study duration.
    - Career pathways, job titles, study durations, and relevant program information.
    - Direct links to faculties and admission requirements for public and private institutions.

3. **Interactive Interest Assessment System:** A multi-dimensional personality and interest test consisting of exactly 42 non-repetitive, scenario-based questions. To ensure maximum diagnostic precision and defensible recommendations, the assessment engine evaluates students across three structured assessment dimensions:

- **Approach 1: Vocational Interests (Holland's RIASEC) - Weight: 1.0:** This approach classifies the student's vocational interest patterns across six personality-environment types (Realistic, Investigative, Artistic, Social, Enterprising, Conventional). It ensures that the recommended major aligns with the student's passions and natural inclinations, promoting long-term academic satisfaction.
- **Approach 2: Core Functional Aptitudes - Weight: 1.5:** As the highest-weighted dimension, this approach measures demonstrated cognitive and problem-solving capabilities. It ensures that the student possesses the required mental stamina and foundational skills necessary to successfully complete the rigorous academic demands of the chosen field.
- **Approach 3: Work Style Preferences - Weight: 1.2:** This approach captures behavioral preferences and environmental compatibility. It ensures that the student is not only capable of studying the major but will also thrive in the typical working environment associated with that career (e.g., preference for teamwork vs. independent work, or office vs. field environments).

4. **Intelligent Tie-Breaker Logic:** The system incorporates a dynamic secondary resolution layer activated when primary dimension scores are highly competitive (within a critical threshold). This ensures precise field assignment by deploying targeted scenarios to transform conflicts into definitive, user-centric and data-driven recommendations.
5. **Intelligent Recommendation Engine:** A logic-based engine that processes user grades, chosen academic fields, and assessment results to generate personalized recommendations. To prevent choice overload, it strictly filters and presents the **Top 3 Recommended Majors**, calculating customized fit percentages, required minimum GPAs, and generating tailored diagnostic reasoning justifying the specific selection.
6. **Exploration and Analysis Tools:** To aid decision-making, the platform includes tools that allow students to explore relevant majors and university programs. This includes structured information on career pathways, program duration, academic requirements, and available study options.
7. **User Personalization and Journey Tracking:** A lightweight user-state system supports a personalized experience. Students can provide profile information, maintain their current session locally, complete assessments, and access generated academic recommendations and improvement plans during their platform journey.
8. **Modern Technical Architecture:** The solution is built upon a highly responsive Single Page Application (SPA) architecture using **Vue.js 3 and Tailwind CSS** for an optimized frontend interface. It utilizes a structured **SQLite** relational database model for efficient data organization and a **Python (FastAPI)** server-side layer to support API operations, data persistence, and recommendation services, laying the groundwork for future AI integration.

1.4 Project Objectives

- **Primary Goal:** To develop an interactive, data-driven digital platform that empowers Jordanian students to make informed, confident decisions about their university majors by scientifically aligning their personal interests, academic grades, and career aspirations.
- **Specific Objectives:**
    - **Personalized & Scientific Guidance:** To develop an intelligent, logic-based assessment system that accurately measures student aptitudes. This includes providing immediate matching for Tawjihi students and generating actionable improvement plans for Grade 9-11 students to bridge skill gaps early.
    - **Centralized Information Hub:** To consolidate dispersed information regarding Jordanian universities, majors, and admission requirements into a single, accessible repository. This eliminates information scarcity and helps students connect academic choices to real-world career pathways.
    - **Decision Quality Improvement:** To reduce the likelihood of unsuitable major selection by facilitating better initial academic decisions. The system aims to minimize the anxiety associated with university applications by replacing social pressure with data-driven insights.
    - **Major Exploration Capabilities:** To enable students to review multiple majors using meaningful criteria such as study duration, required skills, and career pathways, reducing the difficulty of manual information gathering.
    - **Enhanced User Experience:** To create an intuitive, responsive web interface that presents complex academic data in a clear, student-friendly format, ensuring accessibility across diverse learning styles and devices.
    - **Educational Impact & Alignment:** To support Jordan's national vision for educational excellence by equipping students, parents, and academic counselors with reliable tools that enhance overall academic satisfaction and success rates.

1.5 Stakeholders (Beneficiaries) The CareerCompass platform is designed to serve specific groups within the school education sector:

- **Primary Beneficiaries:**
  - **High School Students (Grades 9–12):**
    - **Pre-Tawjihi Students (Grades 9–11):** This group utilizes the platform for early academic planning. They benefit from identifying skill gaps and receiving custom Action Plans to align their abilities with future aspirations before reaching the final year.
    - **Tawjihi Students (Grade 12):** As students preparing for immediate university enrollment, they gain clarity on suitable majors matched specifically to their final stream and grades. The platform empowers them to make confident, data-driven decisions, reducing the risk of poor choices.
- **Secondary Beneficiaries:**
  - **Parents:** Parents gain access to reliable, centralized information regarding market demands and career outcomes, enabling them to provide informed support and understand the rationale behind their child's academic preferences.
  - **Academic Counselors and Teachers:** The platform serves as a supporting utility that reduces administrative workload. Counselors can use the automated assessment reports to provide more qualitative, personalized guidance during advising sessions.
  - **Universities & The Higher Education Sector:** Universities benefit indirectly by receiving applicants who are genuinely suited for their chosen programs. On a national level, this supports educational excellence by potentially reducing major-switching rates and improving student retention.

Chapter 2: System Analysis

2.1 Introduction After identifying the core problem that CareerCompass aims to address and defining the proposed solution, the next critical phase is comprehensive system analysis. This phase involves gathering detailed requirements from stakeholders, understanding data sources and collection methods, and establishing a systematic development methodology.

The system analysis phase provides the foundation for all subsequent development work. Through careful analysis of user needs, functional requirements, data flows, and system constraints, the team establishes clarity on:

- **What the system must accomplish** (Functional Requirements).
- **How the system should behave** (Non-Functional Requirements and System Constraints).
- **Who will use the system and what each user type needs** (User Roles).
- **When each component should be completed** (Timeline and Project Schedule).
- **How the team will organize and manage the development process** (Methodology).

For CareerCompass, we adopted the Agile/Scrum methodology due to its compatibility with evolving user needs, adaptive development processes, and the team's requirement for flexibility as new insights emerge during the development process. This approach aligns with industry standards where projects are impacted by constraints of time and scope, requiring an adaptive and flexible framework to deliver value quickly.

2.2 Data Collection and Requirements Gathering The data collection and requirements gathering process for CareerCompass involved multiple stages, each designed to capture specific information needed to understand system requirements and user needs.

- **Stage 1: Research on Existing Solutions and Market Trends** To establish a baseline for the project, the team reviewed existing academic advising platforms and major comparison tools. This involved analyzing university websites and studying international best practices in academic guidance.
    - **Key Findings:** The research indicated that while international solutions exist, there are limited local platforms providing integrated major selection guidance in Jordan. Most universities still rely on manual advising without digital decision-support tools, confirming a high demand for organized, centralized academic information.
- **Stage 2: Focus Group and User Feedback Research** The team engaged directly with the target audience to validate assumptions. This involved discussions with high school students preparing for university enrollment, as well as academic counselors and teachers familiar with current advising challenges.
    - **Key Feedback:** The feedback highlighted a strong need for data-driven advising tools to support students while reducing the advising workload of counselors and teachers. Students specifically expressed a desire for centralized major information and accessible interest assessment tools.
- **Stage 3: University and Program Information Collection** The final stage focused on populating the system's database with accurate content. The team gathered comprehensive data on university majors, including specific curriculum details, requirements, and expected outcomes.
    - **Data Collected:** This process consolidated essential information such as program descriptions, curriculum maps, career pathway insights, program information, and specific admission prerequisites for Jordanian universities.

2.2.1 Quantitative Analysis: User Questionnaire Findings

To validate the qualitative findings and deeply understand the magnitude of the problem across different educational stages, we designed and distributed targeted online questionnaires. The surveys captured responses from two distinct groups: high school students currently facing the major selection decision (84 responses) and university students who have already experienced the process (31 responses). The analysis of these 115 responses yielded critical statistics that highlight a systemic gap in academic advising.

2.2.1.1 The High School Experience: Anxiety and Lack of Tools The data collected from high school students reveals a high-stress environment exacerbated by a lack of professional support:

- **Fear of Misalignment:** A striking 82.1% of surveyed high school students reported feeling afraid of choosing an unsuitable major that fails to meet their ambitions.

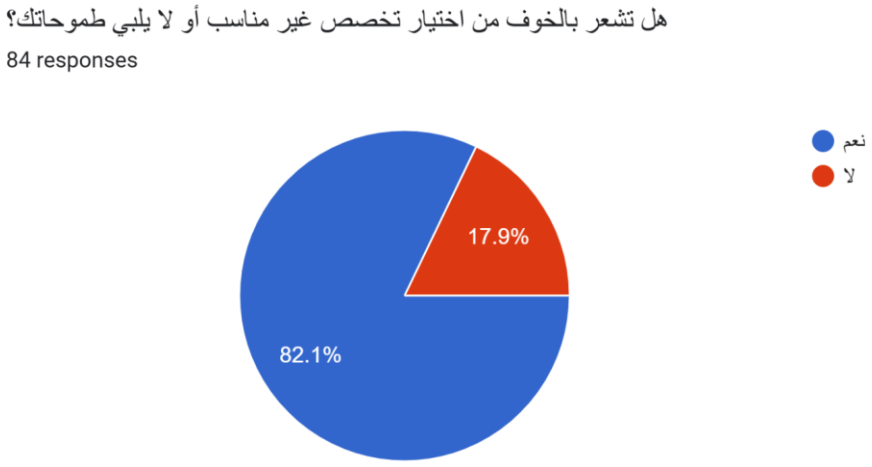


Figure 2.1: High School Students' Fear of Major Misalignment

Figure 2.1 illustrates that a vast majority (82.1%) of surveyed high school students experience significant anxiety regarding their major selection, fearing it will not meet their future ambitions.

- **Demand for Expert Guidance:** Correspondingly, 82.1% of these students expressed a definitive need for a specialist to guide them in selecting their university major.



Figure 2.2: High School Students' Need for Expert Guidance

As shown above, 82.1% of high school respondents expressed a definitive need for a specialized advisor or system to guide them through the complex major selection process.

- **Absence of Assessment Tools**: Despite this high anxiety and demand for help, 76.2% of the high school respondents had never taken an interest or personality test, either in school or online, to help determine their career path. This systemic lack of early diagnostic intervention leaves students navigating one of their most critical life decisions entirely in the dark, relying on guesswork rather than scientifically validated insights.

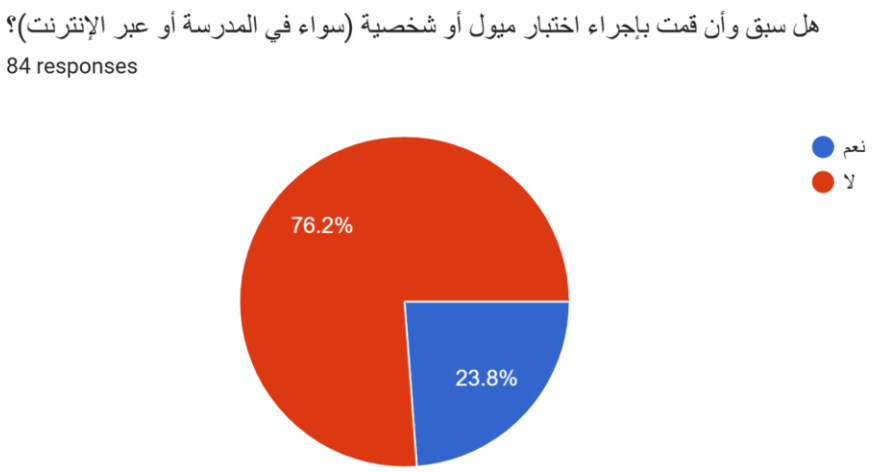


Figure 2.3: Lack of Prior Assessment Tools Among High School Students

Despite their anxiety, 76.2% of high school students reported never having taken an interest or personality assessment to scientifically determine their career path.

2.2.1.2 The University Experience: The Cost of Unguided Decisions The responses from university students illustrate the long-term consequences of entering higher education without prior data-driven guidance:

- **Lack of Prior Guidance:** The lack of early intervention is evident, as 71% of university respondents reported that they did not receive any academic or professional guidance prior to selecting their major. Similarly, 77.4% confirmed they had never taken an interest or personality test before making their choice. This absence of structured pre-enrollment counseling highlights a significant gap in the educational ecosystem, directly contributing to student dissatisfaction.



Figure 2.4: Absence of Prior Academic Guidance for University Students

Among university students, 71% indicated that they received no formal academic or professional guidance prior to selecting their current major.

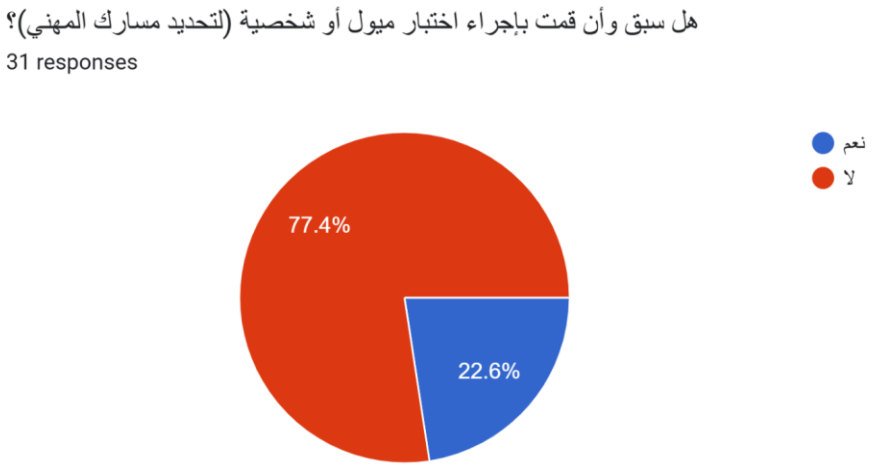


Figure 2.5: Absence of Assessment Testing Prior to Enrollment

Similarly, 77.4% of enrolled university students confirmed making their academic choices without the aid of any psychometric or interest-based testing.

- **Future Uncertainty:** The consequences of this unguided selection persist into higher education, with 67.7% of university students indicating they still feel confused or lack clarity regarding their professional future.

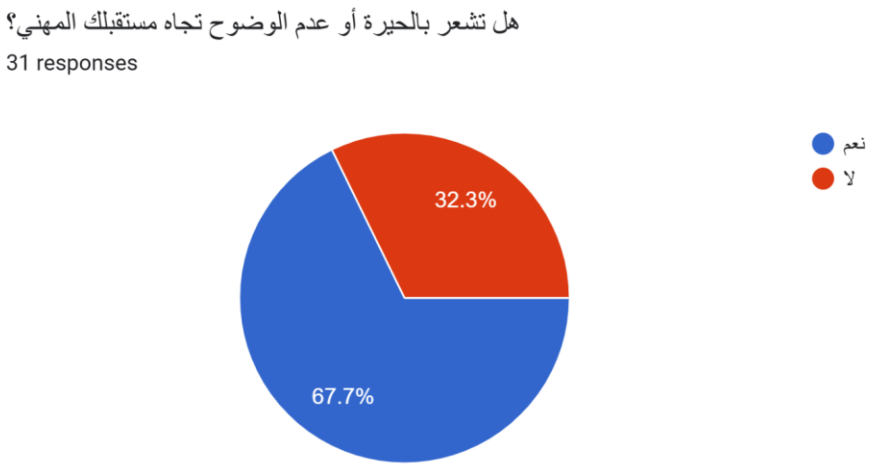


Figure 2.6: Persistent Career Uncertainty Among University Students

The consequences of unguided selection are evident, with 67.7% of university students still feeling confused or lacking clarity regarding their professional future.

- **Major-Reconsideration Concern:** Most critically, 67.7% of surveyed university students reported having considered changing their current major after enrollment. This indicates substantial career uncertainty and highlights the potential consequences of unsuitable initial academic choices.

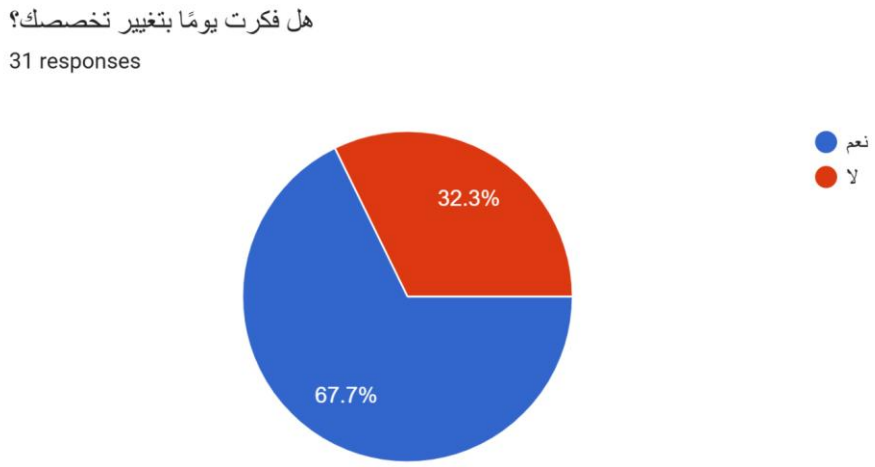


Figure 2.7: Consideration of Major-Switching Among University Students

Critically, 67.7% of surveyed university students admitted to having considered changing their major after enrollment, highlighting the immense cost of early uninformed decisions.

2.2.1.3 Conclusion of Findings These findings strongly support our project's problem statement. The data indicates a clear pattern of inadequate guidance: high school students report anxiety and limited access to assessment tools, while a majority of university students report career uncertainty and have considered changing their major. Although the sample size (115 responses) is limited and non-random, the results justify the need to develop CareerCompass as a dual-track digital advising platform designed to intervene early and provide data-driven recommendations.

2.3 Methodology We adopted the Agile Scrum framework for its iterative and flexible nature. This approach ensures rapid value delivery and effectively manages evolving requirements throughout the development process.

2.3.1 Adapting Scrum for a Small Team While a typical Scrum Team consists of six to ten members, the framework can be scaled for effective use in different team sizes. As a two-member team, CareerCompass adapted the Scrum Core Roles as follows:

- **Product Owner & Scrum Master (Shared Role):** Both members share the responsibilities of the Product Owner, such as articulating customer requirements and maintaining business justification. Simultaneously, they act as the Scrum Master to facilitate the work, remove impediments, and ensure processes are followed.
- **Scrum Team (Development Team):** Both members function as the cross-functional team responsible for understanding requirements and creating the project deliverables (coding, database management, and testing).

2.3.2 Scrum Principles Applied To ensure project success, the team adhered to key Scrum principles:

- **Collaboration:** Constant interaction was maintained to resolve logic issues in the assessment scoring system, ensuring seamless integration and shared ownership of the code.
- **Time-boxing:** Work was divided into short, concentrated work cycles called Sprints. Time was treated as a strict limiting constraint to manage planning, prioritize core features, and ensure effective execution.
- **Iterative Development:** The project was built in sequential increments (e.g., Home Page first, then Test Logic, then University Data) to allow for regular feedback and immediate course correction throughout the lifecycle.

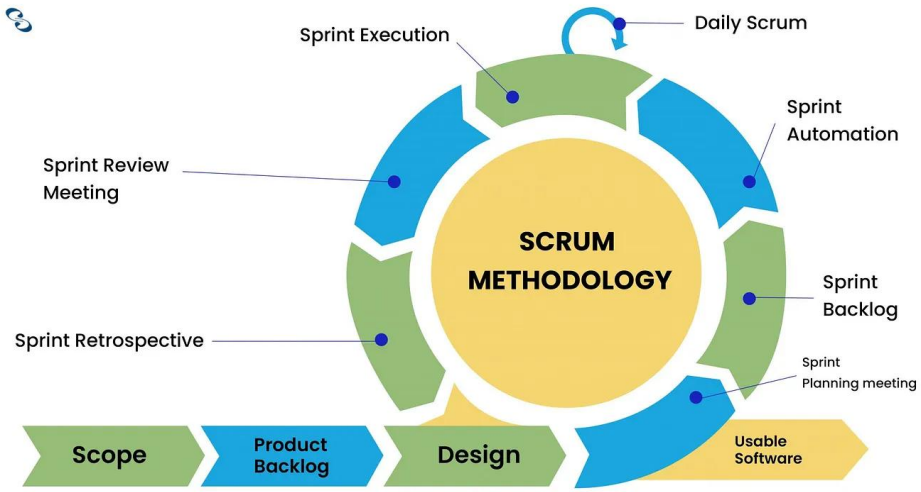


Figure 2.8: The Agile Scrum Methodology Framework

As illustrated in Figure 2.8, the Agile Scrum methodology framework was adopted for the development of the CareerCompass platform. The diagram demonstrates the iterative and incremental software lifecycle, highlighting key phases such as product backlog planning, sprint execution, and continuous feedback loops. This agile approach ensured high adaptability to changing requirements, efficient task management, and the systematic delivery of functional modules at the end of each sprint.

2.3.3 DevOps Integration Practices To enhance the efficiency of our Scrum Sprints and ensure the continuous delivery of a stable web application, the CareerCompass team integrated core DevOps principles into the development lifecycle. DevOps bridges the gap between software development (Dev) and IT operations (Ops), allowing for faster deployment of features and immediate feedback integration.

Our development process followed a continuous iterative cycle tailored to our technical stack:

- **Development & Integration Workflow:**
    - **Plan:** User stories and requirements were selected from the Product Backlog at the start of each 2-week Sprint.

    - **Code:** Development was conducted using VS Code, utilizing **Vue.js 3, Tailwind CSS, and JavaScript** for the frontend and **Python (FastAPI)** for the backend logic.
    - **Repository Collaboration:** Code changes were coordinated through GitHub as the central project repository to maintain version consistency between team members.
- **Local Build & Delivery:**

  ○ **Build & Test:** The application was built and tested locally using Node.js and Uvicorn server environments. Critical components, including assessment scoring and recommendation logic, were verified before each sprint review.

  ○ **Deployment-Ready Assembly:** Instead of a live cloud deployment, the frontend was compiled into a production build and the FastAPI backend was configured to run alongside the SQLite database on a local server. Public cloud hosting is planned as future work.

- **Monitor & Feedback:**
    - **Operate:** The locally running system was monitored for functionality and responsiveness.
    - **Feedback:** Insights gathered from local user testing were fed back into the "Plan" phase of the subsequent Sprint, creating a continuous loop of improvement.

2.4 System Requirements This section outlines the specific requirements necessary for the successful development and operation of CareerCompass. These are categorized into Functional Requirements, which define what the system does, and Non-Functional Requirements, which define how the system performs.

2.4.1 Functional Requirements (FR) Functional requirements define the specific behaviors, functions, and interactions the system must support to meet user needs.

- **User Profile Management:** The system must allow students to create a personal profile using basic information such as name, email, gender, and grade level, without password-based registration.
- **Level Selection:** The system must provide a mechanism for users to select their current academic stage: "Grades 9-11" (Pre-Tawjihi) or "Tawjihi" (Grade 12).
- **Data Input:** The system must accept and validate specific user inputs, including academic grades and major fields (**Health, Engineering & IT, Humanities, and Business**).
- **Assessment Test Logic:** The system must present the interest assessment questions and calculate scores based on predefined logical categories (e.g., **Holland's RIASEC dimensions, Aptitudes, and Work Styles**).
- **Results Generation:** The system must process inputs to display a percentage match for the **Top 3 recommended majors**, along with minimum required GPAs and **tailored diagnostic reasons** justifying each selection.
- **Action Plan Generation:** For Grade 9-11 users, the system must dynamically generate a textual improvement plan identifying skill gaps based on the difference between current grades and future goals.
- **University Search:** The system must provide a searchable and filterable database of localized information regarding Jordanian universities and their programs.

2.4.2 Non-Functional Requirements (NFR) Non-functional requirements specify the quality attributes, performance constraints, and operational standards of the system.

- **Usability:** The user interface must be intuitive, easy to navigate, and culturally appropriate for the target demographic of students aged 14–18.

- **Performance:** The system must optimize logical calculations for test results to ensure they are generated and displayed in under 2 seconds.
- **Responsiveness:** The website design must be responsive, rendering correctly and maintaining functionality across desktop, tablet, and mobile devices.
- **Availability:** The system should target high availability when deployed to a public hosting platform in future phases. In the current local delivery, stable operation was verified during testing sessions.
- **Data Integrity:** The database must ensure that user grades, test results, and profile data are stored accurately and remain consistent throughout transactions.
- **Security:**
    - **Data Privacy & Protection:** The platform follows a frictionless, profile-based model without password authentication and limits data collection to the profile and academic information required for the current workflow.
    - **Access Control via UUIDs:** Each submitted assessment is assigned a non-sequential UUID used to retrieve its stored result. This reduces the risk of simple sequential enumeration but is a lightweight access mechanism rather than full authentication; anyone who obtains a valid UUID may access the associated record.
    - **Input Validation:** All user inputs (such as grades and text fields) must be strictly sanitized and validated at both the frontend (Vue.js) and backend (FastAPI) levels to protect against common web vulnerabilities, such as SQL Injection and Cross-Site Scripting (XSS).
- **Safety:**
    - **Data Recovery:** The system must include backup mechanisms to prevent data loss in the event of a system failure.

- **Fail-Safe Operation:** In the event of an error, the system must fail gracefully, providing user-friendly error messages without exposing sensitive system information or crashing the user session.

2.5 Project Timeline

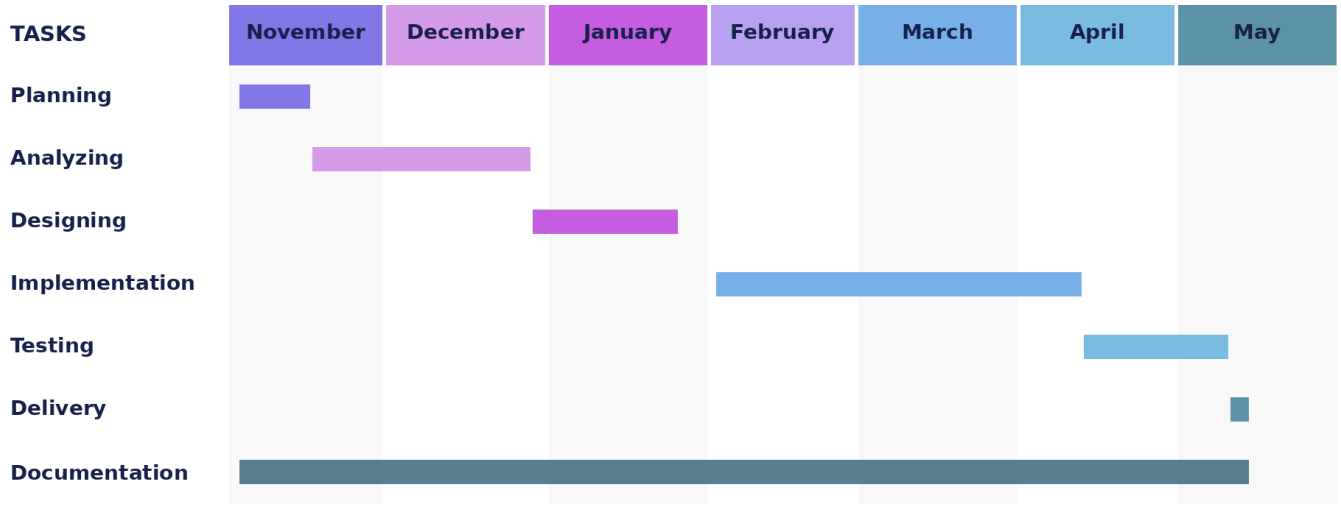
2.5.1 Duration Table

Table 2.1: Duration Table

Tasks	Duration	Start	End
Planning	14 days	2025-11-03	2025-11-16
Analyzing	42 days	2025-11-17	2025-12-28
Designing	28 days	2025-12-29	2026-01-25
Implementation	70 days	2026-02-02	2026-04-12
Testing	28 days	2026-04-13	2026-05-10
Delivery	4 days	2026-05-11	2026-05-14
Documentation	193 days	2025-11-03	2026-05-14

2.5.2 Gantt Chart

Table 2.2: Gantt Chart



2.5.3 Sprints

The project execution was structured using the Scrum framework, adhering to a consistent **2-week time-boxed iteration** schedule. This consistency allows for regular feedback loops and accurate velocity tracking. Complex phases, particularly the implementation of logic and database integration, were decomposed across multiple Sprints to ensure manageability and continuous delivery of value.

Table 2.3: Sprints

Sprint	Dates	Phase	Key Deliverables
S1	Nov 3–16	Planning	Project Proposal, Topic Selection & Approval.
S2	Nov 17–30	System Analysis	Initial Data Collection, Market Trends & Existing Solutions Analysis.
S3	Dec 1–14	System Analysis	Focus Groups, Secondary Data (Jordanian Universities & Fields).
S4	Dec 15–28	System Analysis	Functional (FR) & Non-Functional (NFR) Requirements.

S5	Dec 29–Jan 11	Designing	System Diagrams: ERD (Logical Schema), DFD (Levels 0, 1 & 2).
S6	Jan 12–Jan 25	Designing	UI/UX Wireframes, Database Schema Design (SQLite).
S7	Feb 2–Feb 15	System Implementation	Dev Environment Setup, Vue.js 3 Frontend & Pinia State Management.
S8	Feb 16–Mar 1	System Implementation	FastAPI Backend Initialization, SQLite Database Creation, UUID Logic.
S9	Mar 2–Mar 15	System Implementation	Assessment Interface & Heuristic Scoring Logic (Frontend).
S10	Mar 16–Mar 29	System Implementation	Dual-Track Logic: Recommendation Engine & Gap Analysis (Action Plan).
S11	Mar 30–Apr 12	System Implementation	Frontend-Backend API Integration, Local Server Configuration (Uvicorn).

S12	Apr 13–Apr 26	Testing	Local Unit Testing, Integration Testing, Test Reports.
S13	Apr 27–May 10	Testing	User Acceptance Testing (UAT), Bug Fixing, Performance Optimization.
S14	May 11–May 14	Project Delivery	Final Local Build & System Assembly, Final Documentation Handover.

Chapter 3: System Design

3.1 Introduction

Following the comprehensive system analysis and requirement gathering phases, the project transitions into the system design phase. This stage is critical as it bridges the gap between theoretical requirements and the practical implementation of CareerCompass.

In this chapter, the architectural structure of the web application is defined, focusing on user interaction and data processing. The design process involves creating visual models that represent the logical flow of information and the relationships between data entities. By utilizing Data Flow Diagrams (DFD) and Entity Relationship Diagrams (ERD), a clear blueprint is established to guide the implementation. The design ensures that the system is scalable, efficient, and capable of handling the specific needs of students through a seamless, profile-based interaction model that prioritizes accessibility and ease of use.

3.2 User Interaction and Profile Model

The system is designed to provide a highly accessible, friction-free experience. Rather than requiring traditional, password-protected accounts, CareerCompass utilizes a **Profile-based Model**. This approach allows students to build and manage their academic and career identity dynamically. The platform distinguishes between the following interaction levels:

- **Guest (Visitor): Access Level:** Public.
    - **Capabilities:** Access to landing page content, general information, and university directory browsing. Guests can explore the system's mission but cannot generate personalized diagnostic reports.
- **Student Profile (Pre-Tawjihi / Grades 9-11): Access Level:** Personalized Profile (Planning Track).

    - **Capabilities:** Students create a unique academic profile to record their preferred field and subject grades. This profile serves as a persistent snapshot of their academic progress, allowing them to take assessment tests and receive a personalized improvement plan that identifies skill gaps and focuses areas for future readiness.
- **Student Profile (Tawjihi / Grade 12): Access Level:** Personalized Profile (Decision Track).
    - **Capabilities:** Students manage their profile by inputting final or expected Tawjihi grades. The system leverages this profile data to provide university matching, suggesting the Top 3 recommended majors and universities in Jordan that align with their specific academic standing and personal inclinations.
- **Administrator (Admin): Access Level:** Backend/Data Management.
    - **Capabilities:** Administrative maintenance is performed through backend/database-level operations rather than a dedicated administrator interface in the current version.

3.3 System Design Diagrams

To transform system requirements into a technical reality, it is essential to visualize the logical structure and data movement within CareerCompass. This section presents the architectural blueprints using standard modeling techniques: Data Flow Diagrams (DFD) and Entity Relationship Diagrams (ERD).

These diagrams serve as the bridge between analysis and implementation, providing a graphical representation of how data is processed, stored, and exchanged. This ensures that the backend logic, specifically the recommendation engine and **profile-centric data management**, is structurally sound before implementation.

3.3.1 Data Flow Diagram (DFD)

The DFD serves as a visual representation of information flow. It maps the journey of data, illustrating how user inputs such as academic grades and assessment answers are processed into meaningful outputs like major recommendations. This provides a clear overview of system interactions and the internal processes required to execute the core logic of the platform.

3.3.1.1 Context Diagram

The Context Diagram represents the high-level view of the system boundaries. It depicts CareerCompass as a single process interacting with the Student and the Admin as external entities, highlighting the profile-based data exchange between the user and the system.

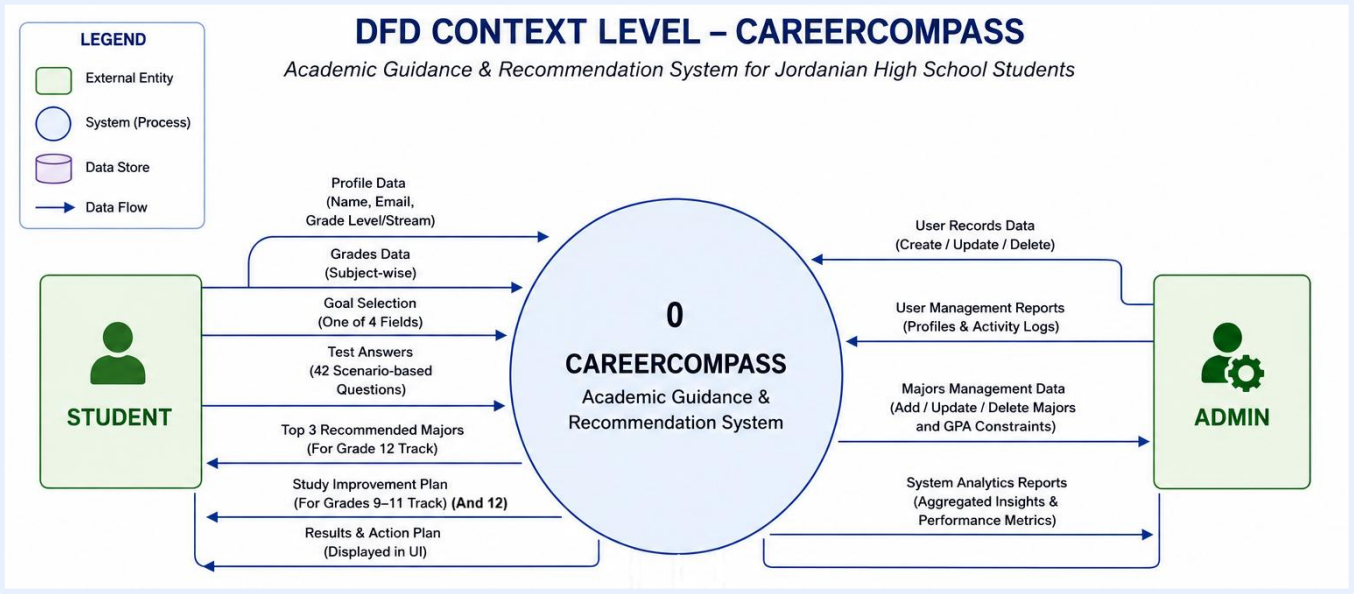


Figure 3.1: DFD Context Level

Figure 3.1 illustrates the Context Level Data Flow Diagram (DFD) for the CareerCompass platform. It presents a high-level overview of the entire system as a single main process, highlighting the primary interactions, inputs, and outputs between the system and its external entities, such as the students and administrators.

3.3.1.2 Level 0 Diagram: The Level 0 Diagram breaks down the system into its core functional components: **Profile Initialization, Academic Data Management, Goal Selection, Interest Test, Recommendation Engine, Study Plan Generator, and Results Presentation**. It highlights the interaction between these primary processes and central data stores, illustrating how student and university data is logically managed and accessed.

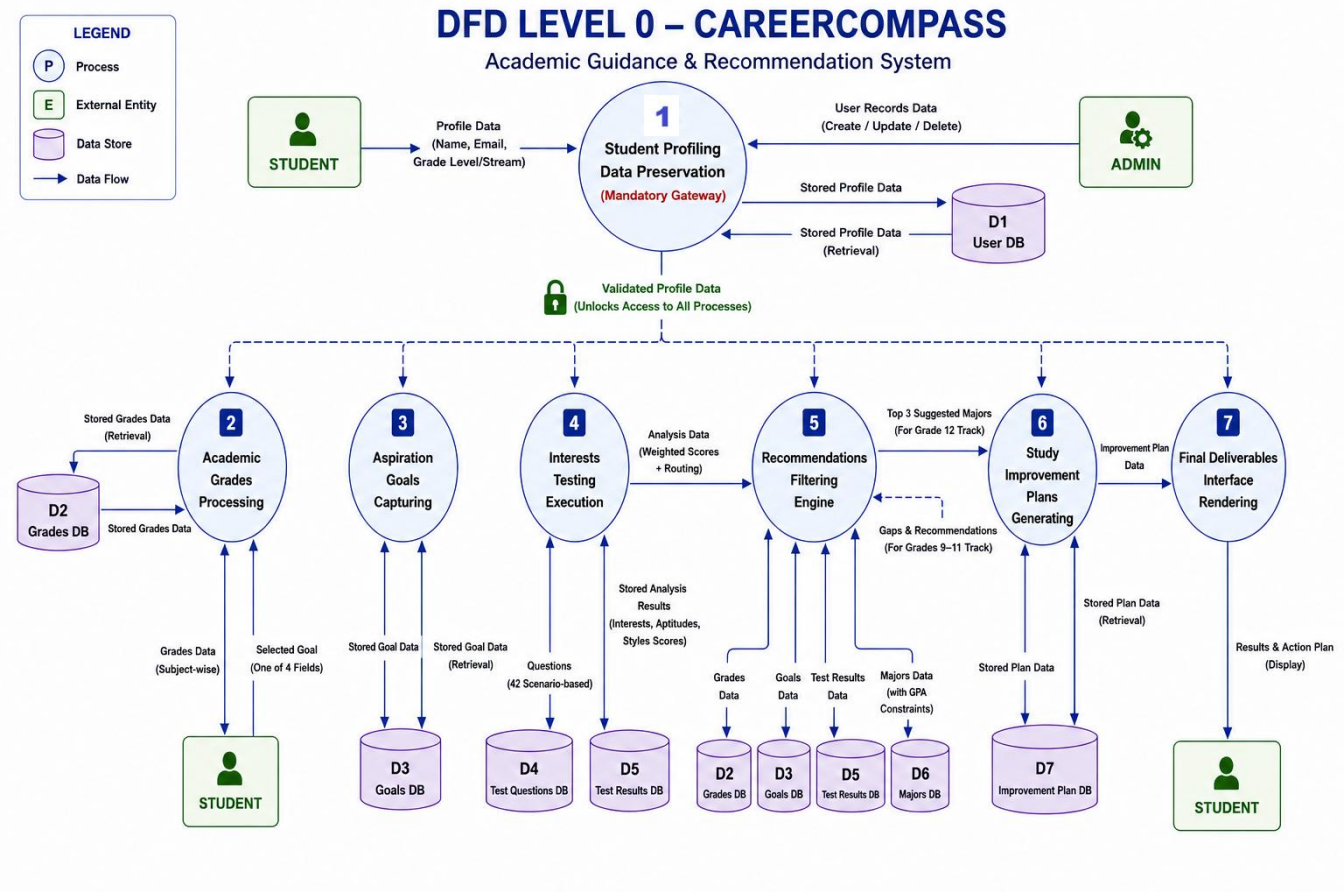


Figure 3.2: DFD Level 0

Figure 3.2 depicts the comprehensive Level 0 Data Flow Diagram (DFD) for the CareerCompass platform. This diagram provides a holistic view of the system's operational logic, illustrating the interaction between seven core functional processes and their associated data repositories. It maps the internal information lifecycle, starting from initial user input through to the final output generation.

The diagram details the precise data movement across the following critical modules:

- **User Profiling & State Management:** Capturing user demographics and managing state persistence without complex credentials.
- **Academic Grades Processing:** Validating and normalizing subject-level scores to ensure accurate admission eligibility calculation.
- **Goals & Interests Capture:** Processing qualitative inputs from the RIASEC assessment and career aspiration modules.
- **Recommendations Filtering:** Utilizing the core recommendation engine to cross-reference personal profiles against dynamic admission datasets.
- **Study Plan Generation:** Transforming academic gaps into structured, actionable, and phased learning roadmaps.
- **Deliverables Rendering:** Finalizing the diagnostic reports and providing actionable UI-driven insights.

By illustrating the bidirectional flow between these processes and the persistent data stores, the DFD serves as a technical proof of concept for the platform's information architecture, ensuring that data integrity is maintained at every stage of the transformation process, thereby supporting a reliable and consistent experience for the end-user.

3.3.2 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) establishes the formal logical structure of the CareerCompass database. It visualizes the core entities such as Profiles, Academic_Grades, Aspirations, and Universities alongside their specific attributes, primary keys, and the complex relationships that bind them.

This diagram serves as the structural blueprint for constructing the underlying SQLite database schema. It goes beyond simple data storage by:

- **Ensuring Data Integrity:** Mapping the cardinality (one-to-one, one-to-many, and many-to-many relationships) between student profiles and their associated academic data, ensuring that every user's progress is uniquely tracked.
- **Optimizing Retrieval:** Defining clear paths for relational queries, which is essential for the rapid extraction of localized Jordanian university information, admission thresholds, and prerequisite subject mapping.
- **Facilitating Scalability:** Providing a robust normalized structure that allows the platform to easily integrate new academic fields or updated university data without requiring significant architectural overhaul.

Ultimately, the ERD acts as the foundation for the system's Backend-to-Frontend communication, ensuring that when a student requests a recommendation, the system can perform complex relational joins efficiently to retrieve the most accurate and up-to-date academic matches, optimizing server response times and significantly enhancing the platform's overall reliability.

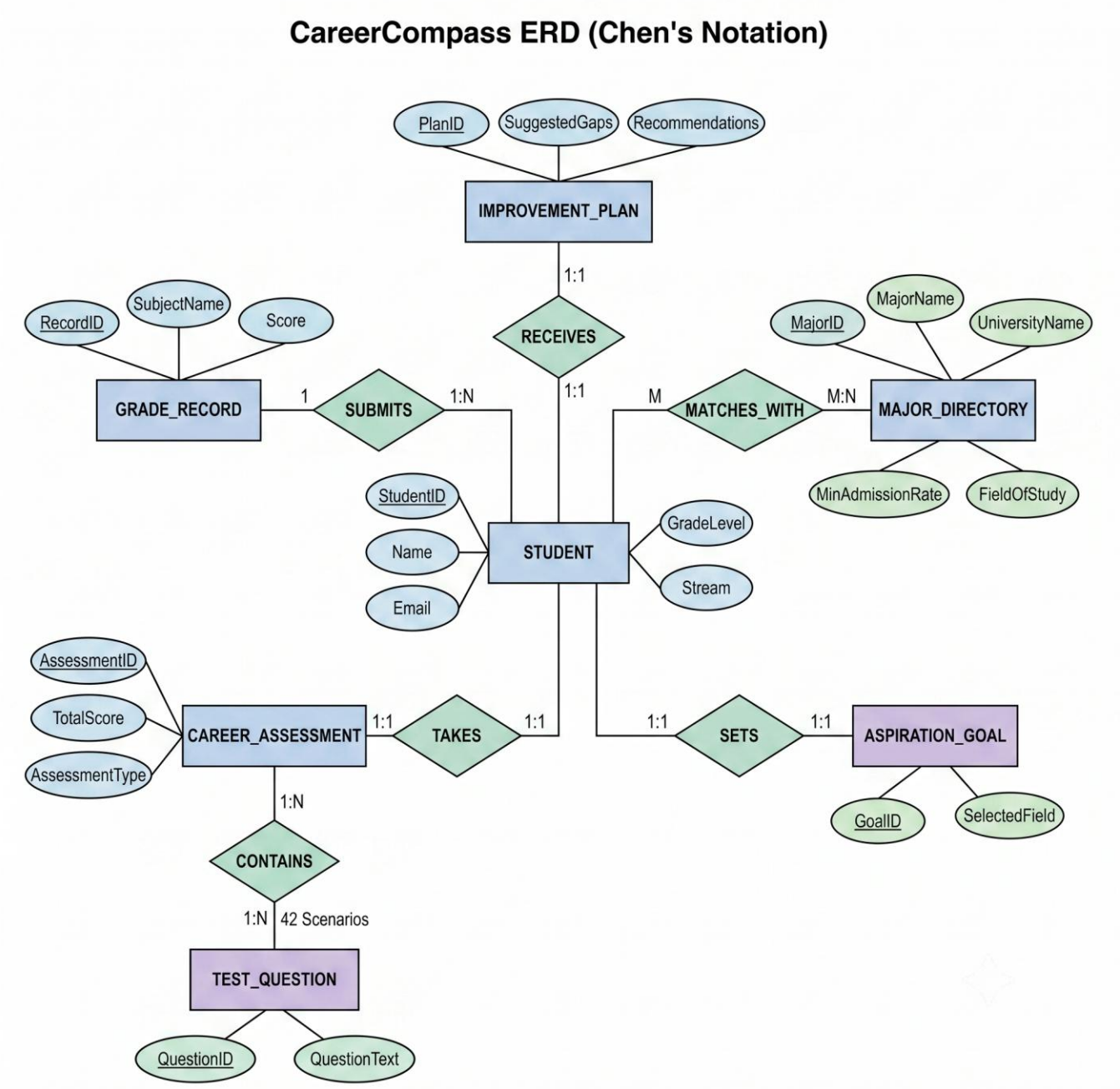


Figure 3.3: CareerCompass ERD (Chen's Notation)

Figure 3.3 illustrates the system's database structure using Chen's ERD notation, mapping the logical relationships between students, their assessments, academic grades, and recommended university majors.

Chapter 4: Software Development and Testing

4.1 Introduction

This chapter provides a comprehensive overview of the implementation phase of CareerCompass. It details the modern frameworks used, the development environment, and the rigorous testing procedures followed to ensure the platform's reliability, accuracy in executing weighted multi-dimensional assessment logic, and efficiency in generating precise academic matching plans.

4.2 Development of the Software The development process followed an Agile Scrum framework, focusing on seamless integration between the modular frontend diagnostic interfaces and the robust server-side processing layer. The system architecture enforces a clean separation of concerns, where the client-side handles dynamic user logic, state reactivity, and immediate UI rendering, while the backend layer manages persistent storage operations and major database queries.

4.3 Programming Languages and Technologies Based on the finalized implementation, the following technologies were utilized:

- **Frontend Framework:** Engineered as a high-performance Single Page Application (SPA) utilizing Vue.js 3 (Composition API) for highly reactive views. JavaScript (ES6+) handles client-side assessment scoring, interaction logic, and validation, combined with Tailwind CSS utility classes to establish a consistent, accessible "Glassmorphism" UI design system.
- **Backend Infrastructure:** Server-side logic and database connectivity are developed using Python within the FastAPI framework, chosen for its high performance and native support for asynchronous API endpoints.

- **Database Management:** SQLite is utilized as the relational database management system to store student/profile data, assessment records, questions, majors, and academic field information.
- **Centralized State Management:** Pinia is integrated directly into the Vue ecosystem to maintain reactive user state, grade profiles, and active assessment workflows seamlessly across route transitions.

4.4 Software Tools

- **Integrated Development Environment (IDE):** Visual Studio Code (VS Code).
- **Repository & Collaboration Platform:** GitHub.
- **Runtime Environment:** Node.js (for the frontend execution) and Uvicorn (as the ASGI server running FastAPI).
- **Interface Design and Prototyping:** Figma.

4.5 Testing and Reviewing the Code Testing was executed iteratively across multiple functional levels to improve system reliability and verify the accuracy of core functionality:

- **Unit Testing:** Thoroughly verified the assessment algorithms responsible for applying multi-scale weights (Interests: 1.0, Work Styles: 1.2, Aptitudes: 1.5) to ensure raw user answers are accurately transformed into valid evaluation dimensions.
- **Integration Testing:** Validated asynchronous requests bridging the Vue.js interface with backend FastAPI endpoints, verifying accurate SQLite database record insertion and reliable querying.
- **Interface and Usability Validation:** Tested responsive Tailwind layouts across desktop screens and mobile viewports, applying careful review to verify consistent Right-to-Left (RTL) formatting and visual readability.

- **Logic Validation:** Manually audited the conditional recommendation engine to confirm that edge-case scenarios trigger the intelligent tie-breaker logic when primary dimension scores show competitive proximity.

4.6 DevOps Principles & Local Build Implementation: In this section, we detail the practical application of DevOps principles that facilitated the continuous development of the CareerCompass platform. While the system operates in a localized environment for this delivery phase, the implementation focused on production-ready technical workflows:

- **Repository & Collaboration:** We utilized GitHub as the central project repository and collaboration platform for coordinating and maintaining code changes. This helped maintain consistency across frontend and backend contributions and provided a shared source for the project throughout the development sprints.
- **Local Integration Workflow:** During implementation, we used Node.js for the frontend and Uvicorn for the backend within the local development environment. Every time a new feature was added, a local build was performed and verified using automated pytest tests alongside manual checks to ensure integration stability before the code was finalized.
- **Deployment-Ready Assembly (Local Build):** Instead of a live cloud deployment, we focused on ensuring the system is strictly "deployment-ready." The frontend SPA was compiled into a final production build, and the FastAPI backend was configured to serve application requests alongside the SQLite database within the local server environment. This approach allowed for rigorous performance monitoring across different emulated devices without incurring cloud hosting overhead.
- **Monitoring and Feedback Loop:** We implemented a continuous feedback loop where insights from local user interface testing were immediately fed back into the development cycle for the next sprint, ensuring the platform evolved according to actual requirements.

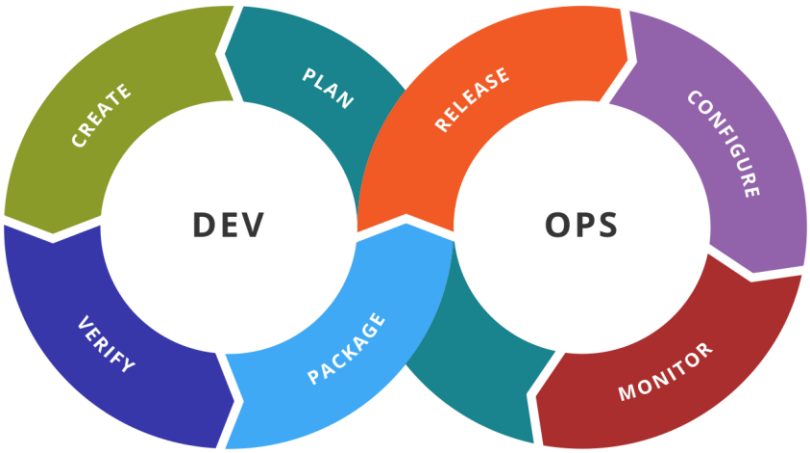


Figure 4.1: DevOps

As illustrated in Figure 4.1, the DevOps lifecycle represents the continuous and iterative methodology adopted during the development of the CareerCompass platform. The diagram outlines the infinite loop of integration between development (DEV) and operations (OPS), encompassing key stages such as planning, creating, verifying, and packaging the software, seamlessly followed by releasing, configuring, and monitoring the system. Adopting this cyclical approach ensured continuous improvement, efficient code management, and overall system stability throughout the project's implementation phases.

4.7 System Interfaces and Results This section presents the graphical user interfaces (GUIs) of the CareerCompass platform and evaluates the system's operational results. The interfaces are designed using a modern tech stack to ensure high responsiveness, user-centric navigation, and seamless visual consistency across all functional modules.

4.7.1 The Home Page Interface The Home Page serves as the primary gateway to the CareerCompass platform. It establishes the visual identity of the system, utilizing an elegant layout featuring a clean background grid and dynamic, animated gradient blobs that elevate user engagement.



Figure 4.2: The CareerCompass Platform Home Page Interface

This interface serves as the main welcome page for the student after creating their profile. Its design is focused on providing a friendly and welcoming environment to make the academic planning process feel less intimidating and more supportive.

Key Functional and Design Aspects:

- **Personalized Greeting:** The page immediately welcomes the student by name. This simple feature makes the platform feel like a personal academic companion rather than a cold, robotic tool, which helps students feel more comfortable using the system.
- **Friendly Visual Design:** The use of an avatar and soft, modern colors is intentional. It aims to create a reassuring atmosphere, helping students feel less stressed about the pressure of university planning and more motivated to start their journey.

- **Ease of Use:** The layout is intentionally kept clean and simple. By focusing on the user's name and a welcoming visual, we ensure that the student's first experience with the platform is positive, encouraging them to explore the various services available on the dashboard without feeling overwhelmed.
- **Clear Purpose:** The text on the page clearly explains the platform's mission: to support the student during the transition from school to university. This helps the user quickly understand the value of the platform and sets the right tone for the academic planning steps that follow.

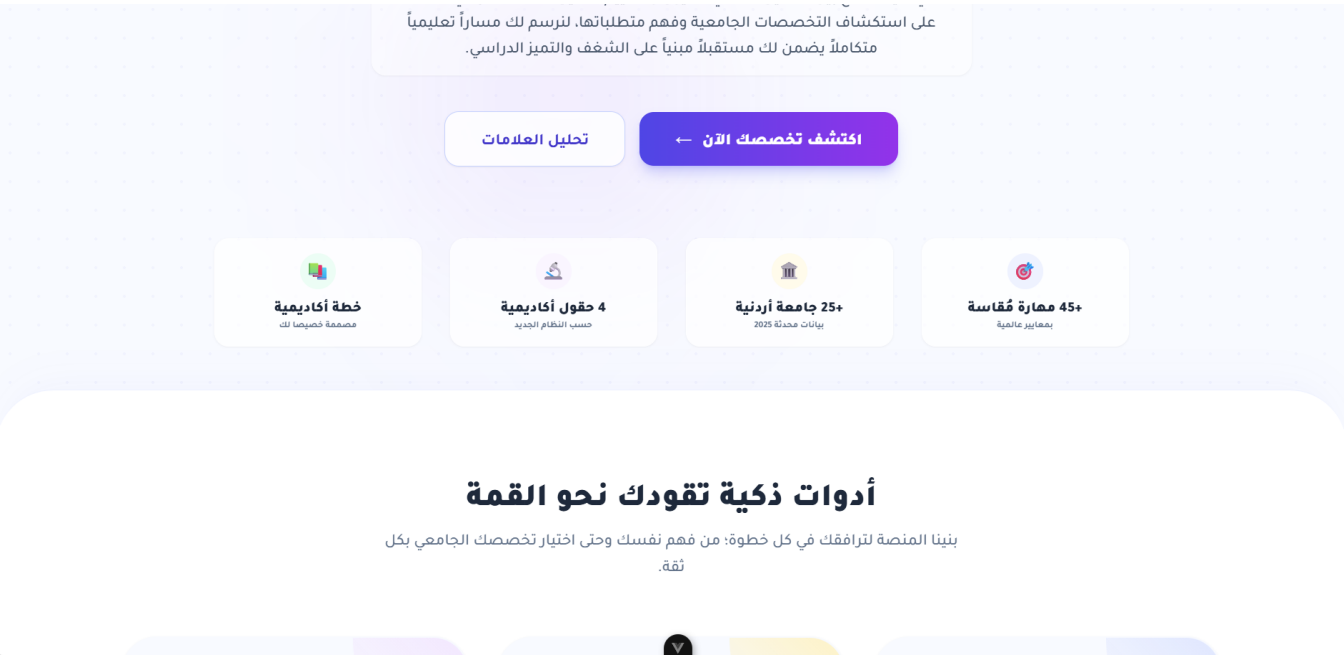


Figure 4.3: Platform Key Features and Statistics Interface

This section of the Home Page highlights the comprehensive scope of CareerCompass, presenting key metrics such as the number of evaluated skills, integrated universities, and academic fields to build user trust and clearly demonstrate the platform's robust, data-backed capacity to handle diverse academic scenarios.

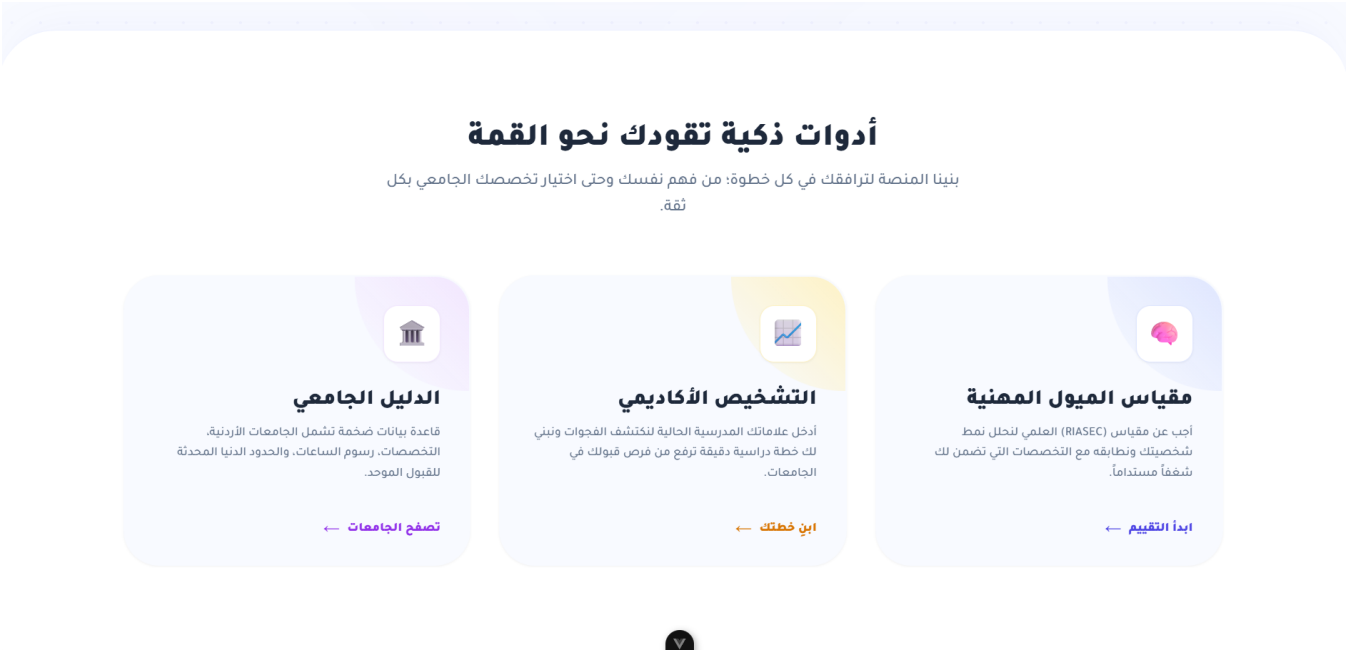


Figure 4.4: CareerCompass Core Services Overview

This interface introduces the platform's three primary tools, the psychometric assessment, the academic diagnostic engine, and the university directory, providing users with direct access to each service.

4.7.2 Student Profiling and Onboarding Interface

To deliver personalized academic guidance, the platform utilizes a streamlined data entry interface to establish the student's session profile. This process is designed to be frictionless, focusing on gathering essential information such as the student’s current academic level, subject preferences, and performance history without the complexity of traditional registration. By creating this dynamic profile, the system immediately aligns its diagnostic engines to the user’s specific context, ensuring that every recommendation, university match, and suggested action plan is tailor-made to support the student’s unique educational goals and future aspirations from their very first interaction with the platform.



Figure 4.5: Student Profiling and Onboarding Interface

This interface captures basic student demographics, including name, email, gender, and academic grade level, initializing a secure user session mapped via a unique identifier.

4.7.3 Multi-Dimensional Career Assessment Interface

This module hosts the core psychometric evaluation tool, utilizing 42 carefully crafted scenarios distributed across three integrated analytical approaches: Interests, Work Styles, and Aptitudes. The design of this interface is intended to minimize user fatigue while maximizing data quality. Each scenario is presented in a clear, interactive format that allows students to express their true inclinations without feeling pressured or overwhelmed by complex jargon.

By gathering input across these three specific dimensions, the platform moves beyond simple career suggestions. Instead, it builds a comprehensive "holistic student profile." This multi-dimensional data is then processed by the internal recommendation engine to find the intersection between what the student loves to do, how they prefer to work, and

what they are naturally capable of achieving. This systematic approach ensures that the final output, be it a major match or a career pathway, is not only based on academic grades but is also deeply rooted in the student's personal identity. Ultimately, this interface transforms the abstract process of career discovery into a structured, reliable, and highly engaging digital experience that provides students with the clarity needed to make informed university decisions.

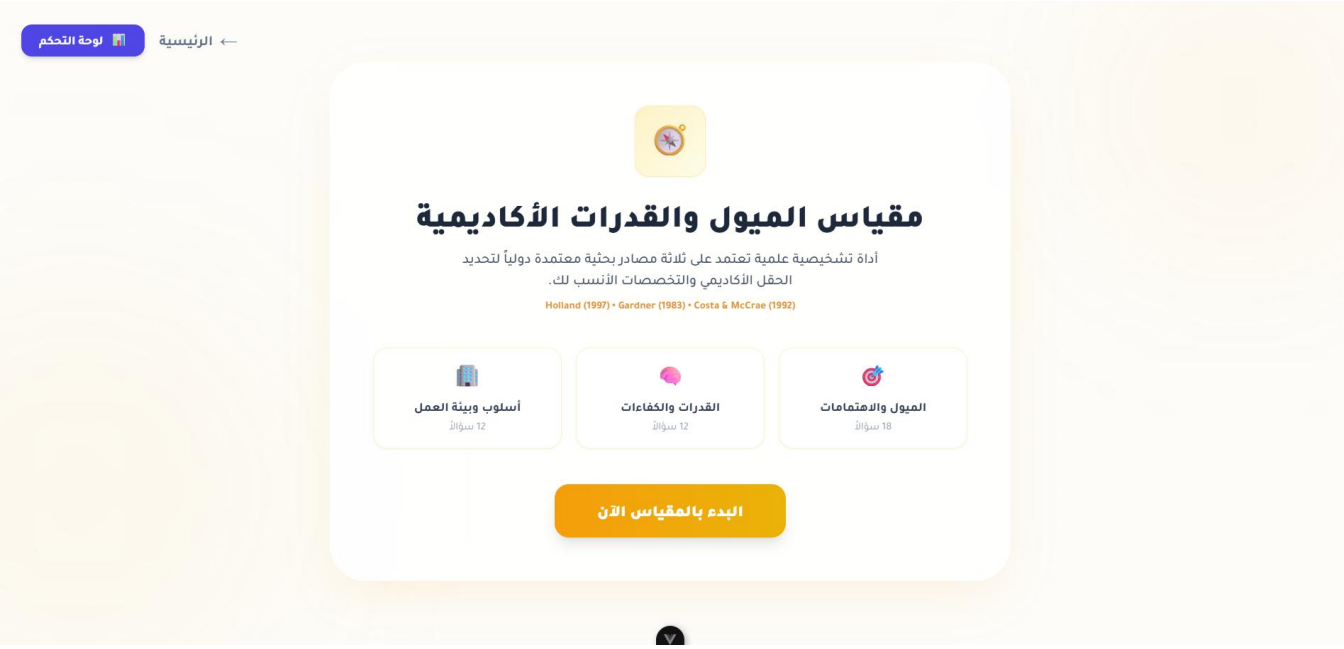


Figure 4.6: Multi-Dimensional Career Assessment Gateway

This landing module introduces the student to the psychometric evaluation process, outlining the three key dimensions evaluated by the 42 scenario-based questions. It clearly explains how these dimensions Interests, Work Styles, and Aptitudes are measured to provide a comprehensive analysis of the student's personal and professional potential, ensuring a clear understanding of the assessment's purpose and its essential role in their final academic journey, thus encouraging thoughtful and highly accurate responses.



Figure 4.7: Psychometric Assessment Results and Field Recommendation Interface

This interface presents the definitive diagnostic outcome generated by the platform's psychometric evaluation engine. It effectively translates abstract assessment scores into actionable academic insights by highlighting the optimal academic field recommendation. Alongside this primary result, the module features a detailed, interactive visual breakdown of the student's RIASEC profile and corresponding behavioral traits. This dual-layered presentation provides a transparent, user-centric, and data-driven rationale for the suggested educational path. By integrating these complex psychometric metrics into a single, intuitive dashboard view, the interface minimizes cognitive overload. It ensures that students can effortlessly grasp their unique academic strengths, natural aptitudes, and work-style preferences. Consequently, users are empowered to confidently align their future educational choices with their personal capabilities and professional aspirations. Ultimately, this seamless synthesis of data serves as a critical bridge between theoretical self-discovery and the tangible reality of university enrollment requirements, simplifying the decision-making journey at this pivotal academic juncture.



Figure 4.8: Comprehensive Psychometric Profile and Field Compatibility Analysis

This interface provides a granular visual breakdown of the student's evaluated RIASEC dimensions and core academic traits. Furthermore, it displays the calculated compatibility percentages across the four primary Jordanian educational fields, offering a transparent, data-driven visualization of the student's inherent strengths.



Figure 4.9: Intelligent Major Recommendation and Diagnostic Reasoning

As illustrated in Figure 4.9, this section presents the definitive output of the recommendation engine, isolating the optimal top three academic majors. Each recommendation features a specific match percentage, the expected minimum admission GPA, and customized diagnostic reasoning that directly links the student's assessed traits to the major's fundamental requirements.

4.7.4 Academic Diagnosis and Grades Input Interface The academic diagnosis component allows the platform to blend scientific interest-profiling with actual scholastic performance.



Figure 4.10: Academic Stage Initialization Interface

This interface initiates the personalized academic diagnosis process. By requiring students to define their current educational stage (Pre-Tawjihi or Tawjihi), the system ensures that the subsequent evaluation logic and generated action plans are accurately tailored to their specific grade level needs and current educational milestones.

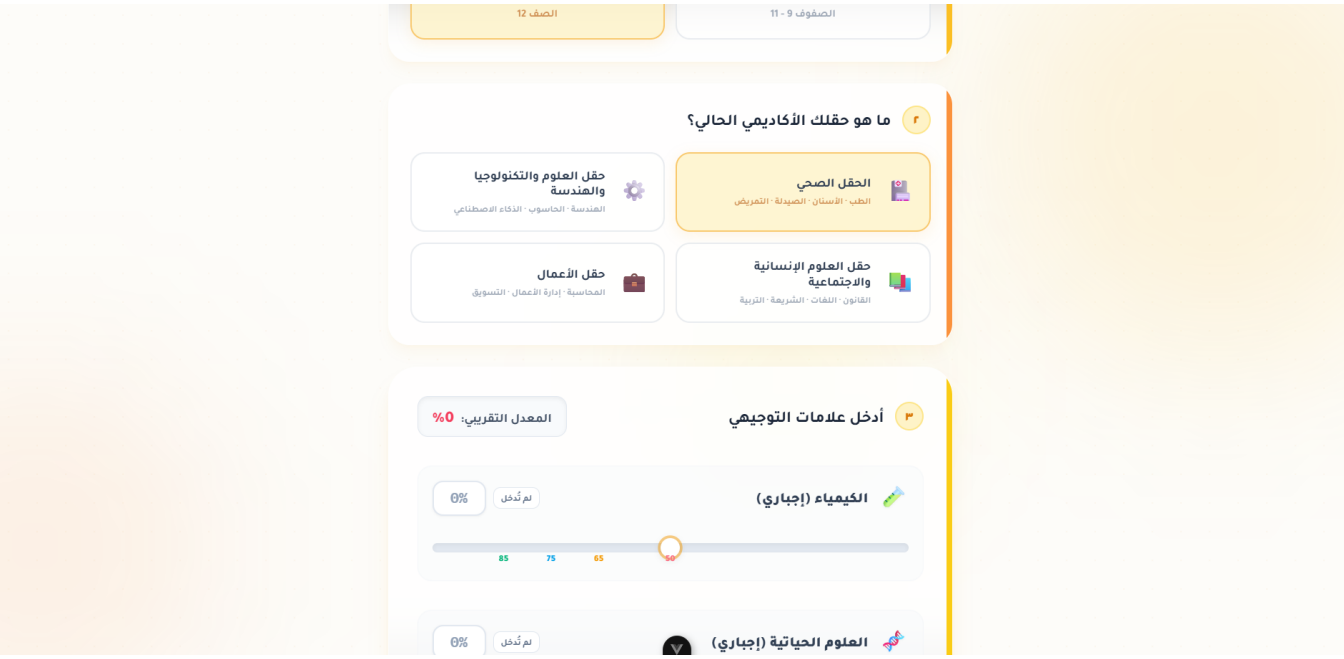


Figure 4.11: Academic Field Configuration and Grade Entry Module

In this module, students select their academic stream (aligned with the 2025/2026 Jordanian curriculum) and input their subject-level scores. This data serves as the foundation for our eligibility and gap analysis algorithms, allowing the platform to generate precise, personalized predictions tailored to each student's chosen path.

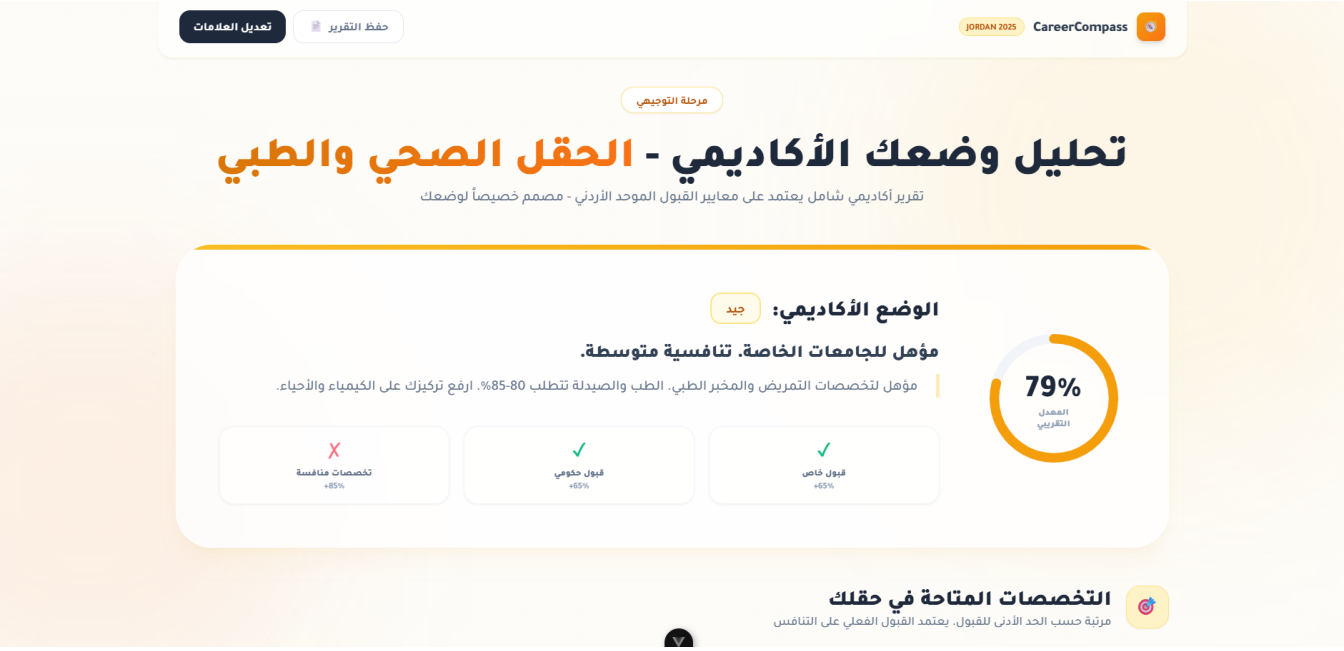


Figure 4.12: Academic Performance and Admission Probability Dashboard

As demonstrated in Figure 4.12, this analytical dashboard calculates the student's indicative GPA and categorizes their competitive standing. It provides an immediate visual assessment based on accurate data regarding their admission probabilities across private, public, and highly competitive university tiers. By converting complex grading data into intuitive graphical insights, the interface empowers students to visualize their academic future and make strategic decisions based on their current performance levels and university admission trends.

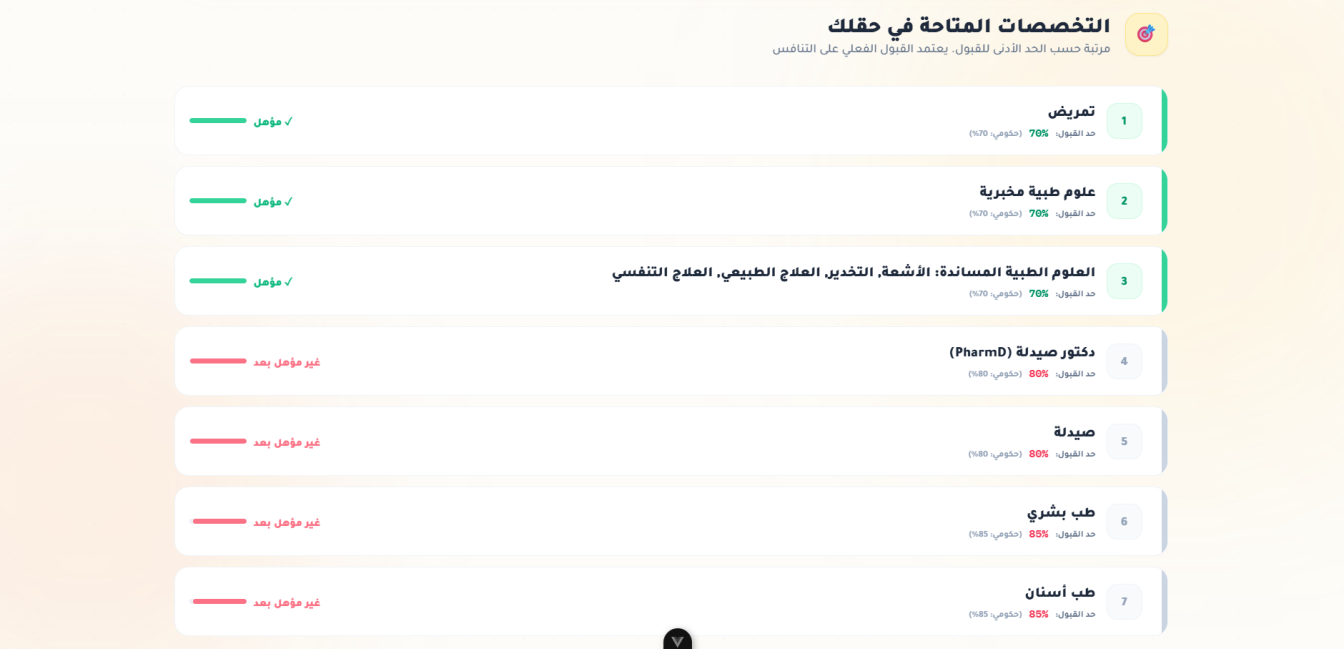


Figure 4.13: Dynamic Major Eligibility and Threshold Filtering

This interface presents a comprehensive, dynamically generated list of academic majors specifically tailored to the student's chosen field. To enhance usability and quick cognitive processing, it employs intuitive, color-coded visual indicators to clearly differentiate between university programs the student currently qualifies for and those requiring further targeted academic improvement based on the latest official admission cut-off rates. By instantly visualizing these eligibility thresholds, the platform empowers students to strategically prioritize their study efforts toward attainable goals.

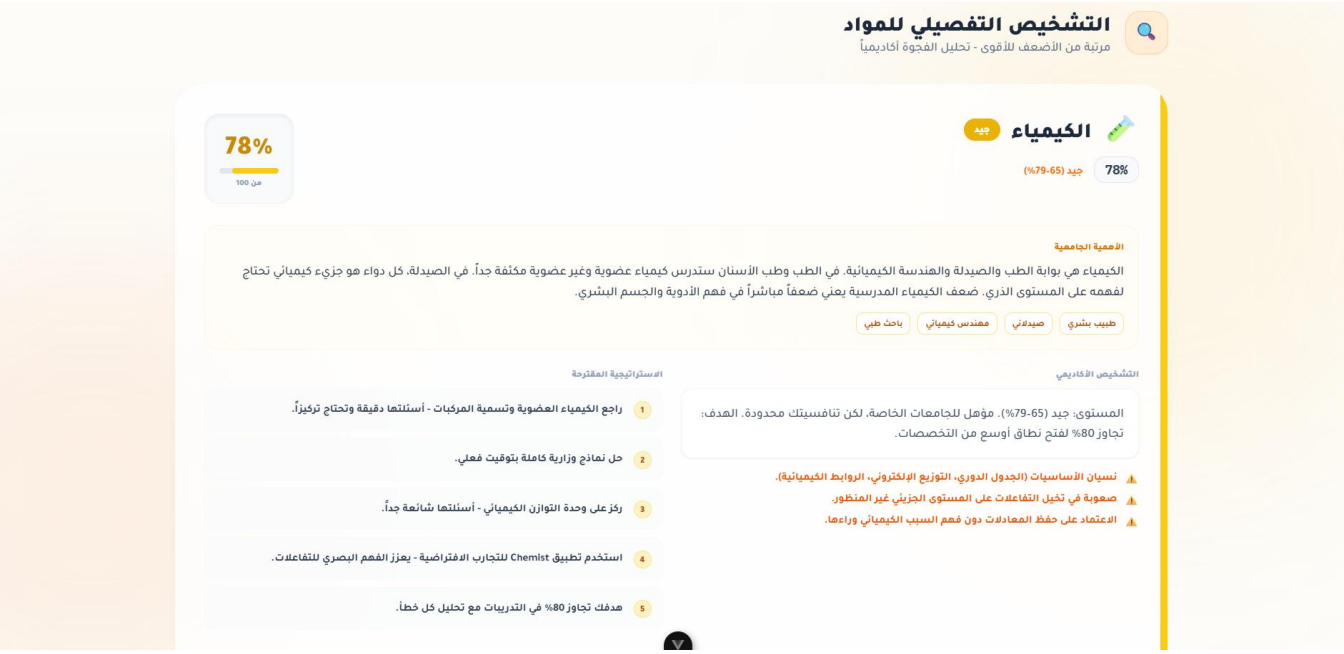


Figure 4.14: Granular Subject-Level Diagnostic Analysis (Chemistry)

This interface demonstrates the system's highly personalized diagnostic capabilities. Rather than providing generic study tips, the engine analyzes the student's exact inputted score for every subject to automatically identify the individualized root causes of academic weakness, such as conceptual misunderstandings versus procedural errors. The platform applies this specialized diagnostic logic across all core subjects, ensuring that each discipline, whether scientific, linguistic, or analytical, receives a bespoke analysis. It dynamically generates a targeted sequential improvement strategy for every module, while explicitly linking the student's mastery in each specific subject to its fundamental relevance in their future university majors. This direct connection significantly boosts intrinsic student motivation by highlighting the practical value of their current educational efforts. By tailoring the feedback to the unique challenges of individual subjects, the system provides a granular roadmap that addresses specific academic deficiencies. Consequently, it guides learners through focused recovery phases, effectively transforming every area of study into a clearly defined pathway toward university excellence and sustained academic success, ensuring they are fully prepared for the rigorous academic demands ahead.



Figure 4.15: Tailored Academic Strategy and Resource Allocation (English)

Showcasing the system's adaptability to diverse academic disciplines, this interface presents a customized diagnostic profile for the English language. It highlights how the platform curates subject-specific external resources ranging from interactive language applications to specialized educational repositories and calculates a realistic timeframe for achieving the target competitive score. By analyzing the student's current proficiency, the system ensures a uniquely crafted roadmap that aligns with their baseline performance. Furthermore, this tailored strategy bridges the gap between language acquisition and practical application, ensuring students recognize how mastering specific linguistic competencies directly supports their performance in advanced university programs. This targeted approach effectively transforms broad academic goals into achievable, time-bound milestones, providing students with the precise resources and strategic clarity required to surpass their current limitations and reach their full academic potential.



Figure 4.16: The Phased "Academic Mastery" Action Plan

This figure illustrates the culmination of the system's gap analysis: a dynamically generated, multi-phase action plan. Designed specifically for the student's calculated performance bracket (e.g., "Competitive Academic Development"), the plan decomposes long-term academic recovery into manageable, time-boxed phases (e.g., "Closing Detailed Gaps" and "Building Accuracy and Speed"). It provides actionable, metric-driven steps to ensure measurable and highly personalized academic progression.

4.7.5 Jordanian Universities Directory This interface serves as an integrated information hub containing comprehensive, up-to-date data regarding local higher education options. It acts as a centralized source that consolidates institutional details, faculty structures, and program-specific requirements for all Jordanian universities. By providing this consolidated data, the platform ensures that students have access to a reliable and inclusive guide, allowing them to explore academic opportunities and program details in one place. This transparency simplifies the research process, helping students make informed and confident decisions about their future university path without the need to search multiple external sources.

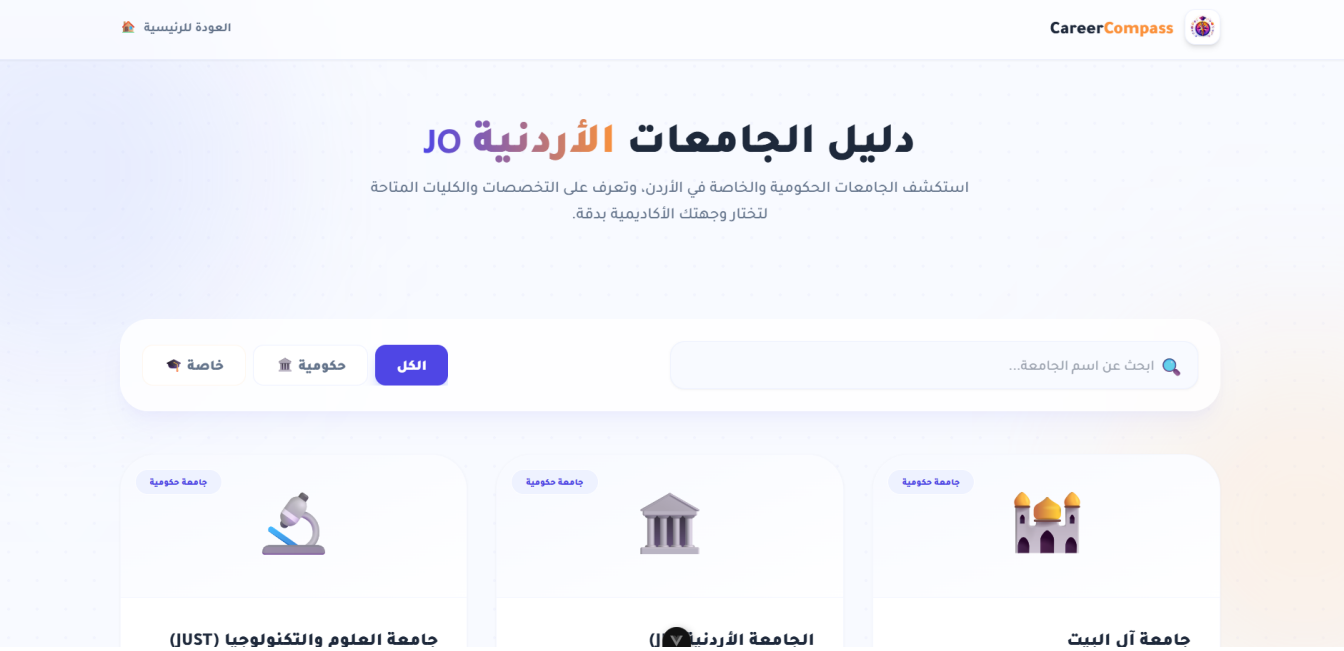


Figure 4.17: Jordanian Universities Directory - Search and Filter Interface

This interface serves as the centralized entry point for the universities directory. It features an intuitive search mechanism and advanced dynamic filtering capabilities, specifically categorizing public versus private institutions. This functionality allows students to efficiently navigate and explore the comprehensive repository of localized higher education institutions. This targeted design ensures that users can quickly narrow down their search parameters to find relevant universities that directly align with their personal preferences and future academic goals without unnecessary navigation steps. By mitigating information overload and centralizing scattered institutional data, this streamlined gateway significantly accelerates the research phase. Consequently, it enables students to evaluate their prospective academic destinations with greater accuracy, clarity, and absolute confidence. This ultimate transparency ensures that every final educational choice is backed by reliable institutional facts rather than informal advice or external social pressure.

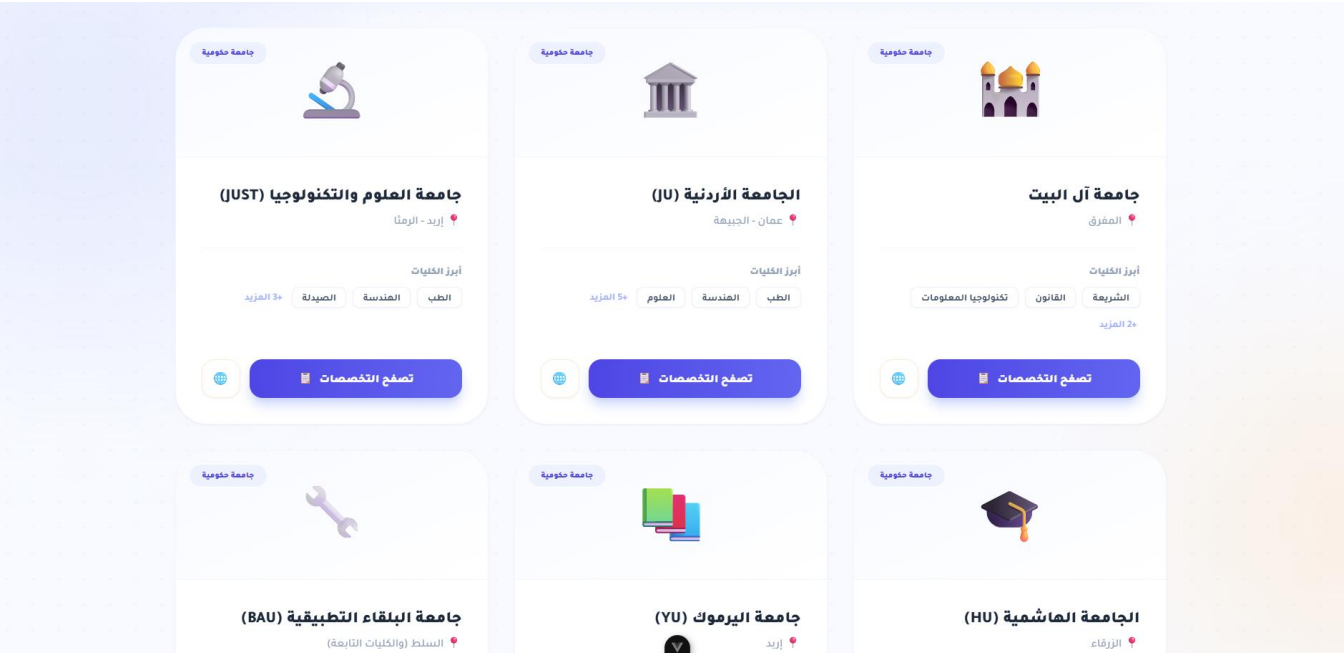


Figure 4.18: University Profiles and Faculty Overview Grid

This section displays the filtered university profiles through an organized grid layout designed to maximize readability and reduce visual clutter. Each institutional card provides essential primary data at a glance, including the geographical location, official university emblems, and a snapshot of prominent academic faculties. This specific setup serves as a direct navigation portal to detailed program specifications, guiding users smoothly and significantly reducing the number of required clicks. By presenting this complex information in a structured visual format, the platform helps students compare multiple institutions simultaneously with exceptional clarity. This intuitive presentation enables a much faster and more effective research experience when exploring their potential future university options, ultimately empowering students to make highly informed educational decisions without feeling overwhelmed by scattered data.

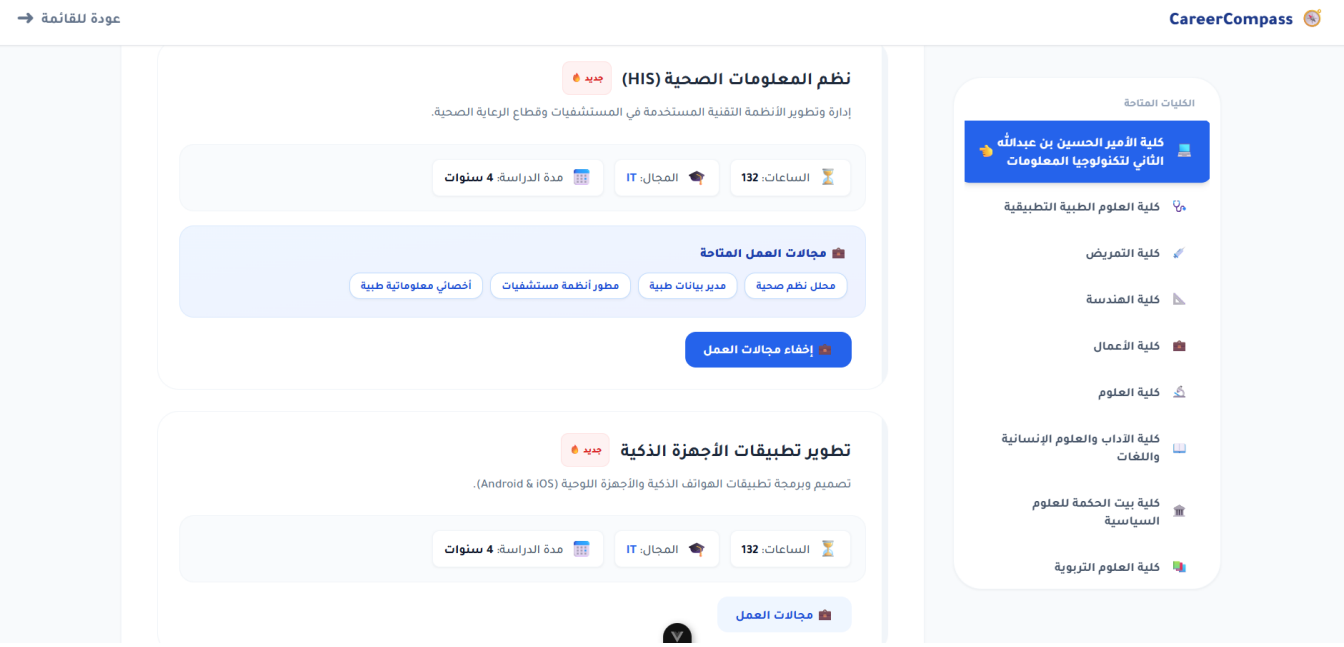


Figure 4.19: Detailed Faculty Programs and Career Pathways Interface

Upon selecting a specific university faculty, this granular interface presents the available academic programs. It details essential metrics such as total credit hours, study duration, and the overarching academic field. Furthermore, it incorporates an interactive toggle to reveal specific career opportunities and job market pathways associated with each major, directly linking academic choices to professional outcomes.

4.7.6 Student Dashboard and Improvement Plan The Student Dashboard represents the ultimate outcome phase of the CareerCompass ecosystem, where analytical backend computations are translated into actionable insights. This interface serves as the primary hub for the student's academic journey, presenting a highly personalized diagnostic report that evaluates their current performance against official admission benchmarks. By consolidating complex data such as identified academic gaps, subject-specific analysis, and admission eligibility probabilities, the dashboard transforms abstract grading metrics into a structured, clear roadmap for the student. This presentation of information empowers students to understand their academic standing clearly, providing them with a concrete and simplified plan of action to focus their efforts effectively. By delivering this guidance in a

clear, consolidated view, the dashboard ensures that students receive a comprehensive summary of their academic status, enabling them to make well-informed decisions regarding their future university path.

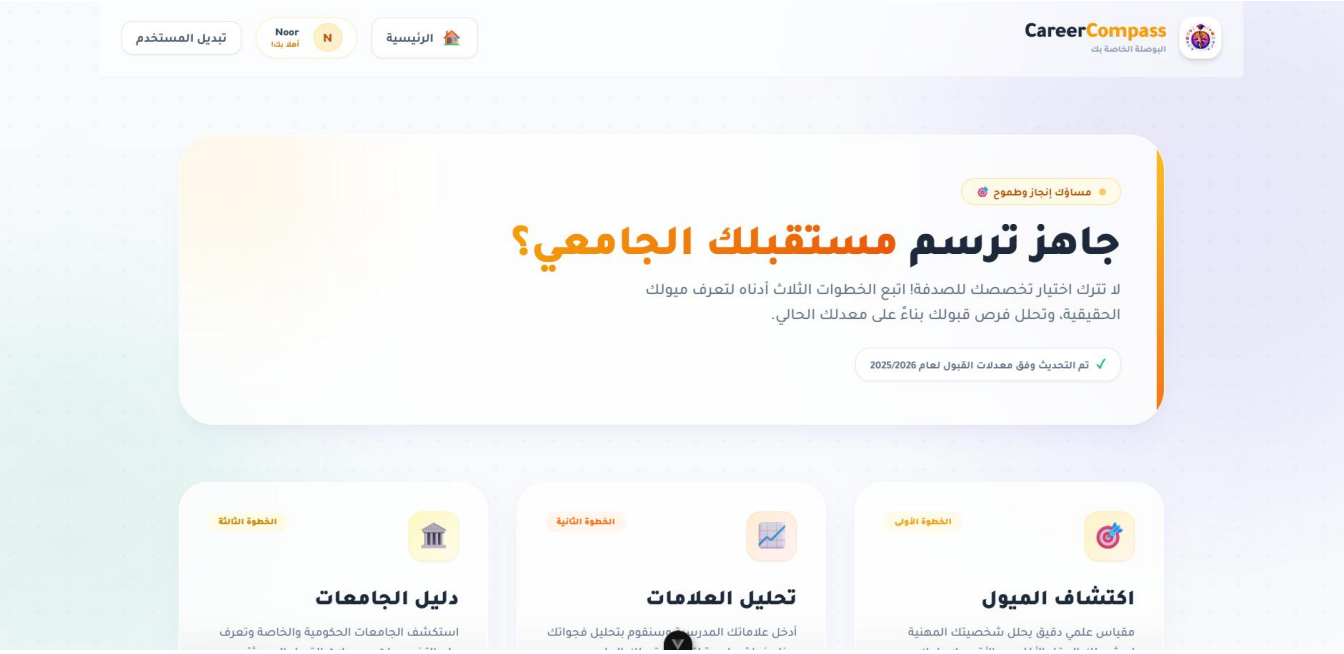


Figure 4.20: The Main Student Dashboard and Navigation Hub

This interface serves as the primary navigation hub for students with an active profile. It provides a clear, centralized entry point to the system's three core services: the psychometric interest assessment, the academic grade analysis, and the university directory. Additionally, the dashboard displays a status indicator confirming alignment with the latest 2025/2026 admission thresholds.

4.8 Summary

In conclusion, this chapter detailed the implementation phase, technical stack, and testing procedures for CareerCompass. The presented interfaces demonstrate how the system’s core algorithms and multi-dimensional logic are translated into an intuitive, seamless experience for students, ensuring the platform effectively delivers actionable academic guidance.

Chapter 5: Challenges and Difficulties Encountered

5.1 Introduction Developing an effective academic guidance system requires highly stable underlying data. However, during the software development lifecycle of CareerCompass, complex systemic and technical obstacles were encountered. This chapter discusses the core difficulties faced, primarily driven by the dynamic and shifting nature of the Jordanian educational landscape, alongside advanced client-server integration hurdles.

5.2 The Shifting Landscape of the Jordanian Tawjihi System The most formidable administrative challenge encountered during system design and database population was the inherent instability of the Jordanian secondary education system (Tawjihi). The Ministry of Education frequently implements structural updates, requiring continuous algorithmic and architectural adaptation:

- **Frequent Changes in Academic Fields and Prerequisites:** The constant updates from the Ministry of Education made it highly challenging to maintain a stable system. The platform was initially designed and programmed based on a six-field educational framework. However, sudden official decisions reduced this structure to only four fields. This major shift forced the development team to completely rebuild the core logic of the recommendation engine. Furthermore, the continuous alteration of mandatory prerequisite subjects required repeated rewriting of calculation codes, adjustments to the gap analysis formulas, and extensive database updates to ensure alignment with the latest regulations.
- **Inter-Generational Shifts (Cohort Variations):** Requirements and passing criteria in Jordan often shift from one academic generation (Cohort) to the next. Different cohorts frequently face altered admission policies, dynamic prerequisite subjects, and fluctuating minimum acceptance GPAs. Designing a database schema

flexible enough to evaluate students accurately based on their specific cohort rules was highly complex.

- **Curriculum Updates and Subject Weighting:** The frequent introduction of new foundational subjects and ongoing changes to how individual subject grades are weighted to calculate the final Tawjihi average demanded a highly adaptable calculation engine capable of accommodating real-time formula adjustments.
- **Data Scarcity and Fragmentation:** Because ministerial decisions and specific university admission thresholds for newly introduced fields are updated dynamically, gathering consolidated, localized, and verified database entries proved exceptionally difficult and required continuous manual validation.

5.3 Algorithmic and Technical Integration Challenges Beyond systemic data challenges, bridging the advanced assessment logic with modern web technologies presented significant technical hurdles:

- **Multi-Model Weight Balancing:** Fine-tuning the mathematical integration between Holland's RIASEC professional interests, cognitive aptitudes, and work style stamina to output a singular, optimal field recommendation required extensive testing to prevent logical bias.
- **Intelligent Tie-Breaker Implementation:** Developing a secondary dynamic resolution layer to handle edge cases where primary dimension scores resulted in a tie required complex conditional logic to seamlessly trigger student-driven scenarios without breaking the assessment flow.
- **Asynchronous Flow Management:** Maintaining multi-step assessment questionnaires and persistent student grade context inside client-side Pinia stores smoothly while preventing UI freezing during asynchronous backend API fetches required meticulous state management and error handling.

Chapter 6: Conclusion and Future Work

6.1 Conclusion The CareerCompass platform successfully addresses the acute pre-enrollment academic guidance gap within the Jordanian educational framework. By synthesizing psychological interest assessment with structured academic performance metrics, the project delivers a highly accessible digital advisory utility. Implementing custom multi-track logic successfully supports both early preparation via gap-analysis action plans for Grades 9–11 and immediate, data-driven "Top 3 Major" matching aligned directly with Jordan's new secondary fields for Grade 12 students. Ultimately, the system demonstrates how modern web architectures can support students in making informed career decisions while potentially reducing problems associated with unsuitable major selection.

6.2 Future Work To expand the platform's technological capabilities and educational scope, several strategic enhancements are outlined for future iterations:

- **Cloud Deployment and Public Hosting:** Transitioning the locally built system to a robust cloud infrastructure (such as Netlify for the frontend and cloud-managed servers for the FastAPI backend) to ensure continuous public availability and scalable performance for students nationwide.
- **Secure Authentication:** Adding password-based or OAuth authentication for stronger profile and assessment access control.
- **Generative AI Conversational Assistant:** Integrating an advanced Large Language Model (LLM) API to serve as an intelligent, round-the-clock interactive guidance chatbot answering custom student queries.
- **Unified Admissions System Integration:** Developing dynamic backend bridges to sync directly with Jordan's national unified admissions portal to present real-time seat availability and updated cut-off GPAs.

- **Native Mobile Application Releases:** Packaging the responsive web application into dedicated, standalone iOS and Android releases to provide enhanced offline access and continuous push notifications.
- **Comprehensive Longitudinal Tracking:** Implementing multi-year academic portfolio tracking to monitor skill development from Grade 9 through successful university graduation.

Sources

[1] Project Management and Agile Methodologies

- **SCRUMstudy.** (2022). A Guide to the Scrum Body of Knowledge (SBOK™ Guide) (4th ed.). VMEdu, Inc.

[2] Systems Analysis and Database Design

- **Kendall, K. E., & Kendall, J. E.** (2019). Systems Analysis and Design (10th ed.). Pearson. (Referenced for DFD and System Modeling).
- **Elmasri, R., & Navathe, S. B.** (2015). Fundamentals of Database Systems (7th ed.). Pearson. (Referenced for ERD Chen's Notation and Database Relational Design).

[3] Software Development & DevOps Principles

- **Bass, L., Weber, I., & Zhu, L.** (2015). DevOps: A Software Architect's Perspective. Addison-Wesley Professional.
- **Kim, G., Debois, P., Willis, J., & Humble, J.** (2016). The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations. IT Revolution Press.

[4] Academic Data and Jordanian Educational Regulations

- **Ministry of Higher Education and Scientific Research (MOHE), Jordan.** (2024). Unified Admission Coordination Unit (UACU) Data and Cut-off GPAs. Retrieved from http://www.admhec.gov.jo
- **Ministry of Education (MoE), Jordan.** (2024). Secondary Education (Tawjihi) New Academic Streams and Regulations. Retrieved from https://moe.gov.jo

[5] Psychometric and Career Assessment Modeling (The 3 Approaches)

- **Approach 1 (Vocational Interests):** Holland, J. L. (1997). Making Vocational Choices: A Theory of Vocational Personalities and Work Environments (3rd ed.). Psychological Assessment Resources (PAR).
- **Approach 2 (Cognitive Aptitudes & Skills):** Fleishman, E. A., & Reilly, M. E. (1992). Handbook of Human Abilities: Definitions, Measurements, and Job Task Requirements. Consulting Psychologists Press.
- **Approach 3 (Work Styles & Environment):** National Center for O*NET Development. (2024). O*NET Work Styles and Values Framework. U.S. Department of Labor. Retrieved from https://www.onetonline.org

[6] Programming Frameworks and Technologies

- **Vue.js Core Team.** (2024). Vue.js 3 Documentation: The Progressive JavaScript Framework. Retrieved from https://vuejs.org
- **Ramírez, S.** (2024). FastAPI Documentation: High-performance, easy to learn, fast to code, ready for production. Retrieved from https://fastapi.tiangolo.com
- **Tailwind Labs.** (2024). Tailwind CSS Documentation. Retrieved from https://tailwindcss.com
- **Hipp, R. D.** (2024). SQLite Database Engine Documentation. Retrieved from https://www.sqlite.org

[7] Educational Psychology and Academic Improvement Models

- **Bloom, B. S., Krathwohl, D. R., & Anderson, L. W.** (2001). A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom's Taxonomy of Educational Objectives. Longman. (Referenced for the categorization of academic errors: Conceptual vs. Procedural Knowledge).
- **Brown, P. C., Roediger III, H. L., & McDaniel, M. A.** (2014). Make It Stick: The Science of Successful Learning. Harvard University Press. (Referenced for the

"Academic Mastery Program" strategies, including retrieval practice and high-pressure testing simulations).

- **Black, P., & Wiliam, D.** (1998). Assessment and Classroom Learning. Assessment in Education: Principles, Policy & Practice, 5(1), 7-74. (Referenced for the foundational logic behind the Formative Gap Analysis and Action Plan generation).